

MAINTENANCE OF PERIOPERATIVE THEATRE INSTRUMENTS AND APPARATUS

1.1 Introduction to the unit of learning

This unit standard specifies the competencies required in maintaining perioperative theatre instruments and apparatus. It involves necessary precautions importantly followed in the context of operation theatre setting to avoid breakage, accidents, and infection. It also includes general repair of Operation theatre instruments and apparatuses

Upon completion of this unit the learner will be able to prepare to dis-assemble and assemble perioperation theatre instruments, check and assemble perioperation theatre instruments, pack and sterilize perioperation theatre instruments, store operation theatre instruments and apparatus, evaluate care of instruments and apparatuses in Perioperation theatre.

Learning Outcome 1: Prepare to dis-assemble and assemble Perioperation Theatre instruments

Introduction

This unit trainees will train how to adhere to safety practices in Operation Theatre, tools for dis-assembling and assembling theatre instruments and apparatus, classification of surgical instrument and Operation Theatre Practice

Definition of key terms

Alloy: A combination of several different kinds of metals. Alloys are used in the manufacture of stainless steel.

Box lock: The hinge point of many surgical instruments.

Chisel: An orthopedic instrument used to slice bone; one side is straight and the other is beveled.

Curettage: The removal of tissue by scraping with a surgical curette.

Dilator: Graduated, smooth instrument that is used to increase the diameter of an anatomical opening in tissue.

Double-action rongeur: A bone-cutting instrument with two hinges in the middle. This increases leverage and strength of the instrument.

Elevator: A straight instrument with curved sharp or dull tip used to separate tissue layers such as periosteum from bone.

Gouge: A V-shaped bone chisel. **Hemostat:** A surgical clamp most often used to occlude a blood vessel.

Rongeur: A hinged instrument with sharp, cup-shaped tips that is used to extract pieces of bone or other connective tissue. **Serosa:** The delicate outer layer of tissue of most organs.

Anodized: The instrument is exposed to conditions that cause an oxide coating that is relatively impenetrable to atmospheric oxygen. Instruments can be anodized to reduce reflections. **Atraumatic:** Without injury.

Crushing: Destructive effects of specific instruments. Some procedures require the use of crushing clamps. **Cutting Separating** with a sharp instrument or device.

Debulking: Decrease in mass or volume using an instrument or device.

Dilation: Enlarging an opening in a progressive manner.

Dissection: Process of separating tissues through anatomic planes by using sharp or blunt instrumentation.

Evacuating: Emptying a cavity or space.

Grasping Holding in a traumatic or atraumatic manner.

Instillation: Fluid is slowly introduced into a cavity or space.

Occlusion: **Closing** a lumen for the purpose of the procedure. The closure can be permanent or temporary.

Percutaneous: Enter directly through the skin; without incision.

Retraction: Stabilizing a tissue layer in a safe position for exposure of a part. A retractor can be manual or self-retaining.

Shank: The area of a surgical instrument between the box lock and the finger ring. **Single-action rongeur:** A heavy cutting instrument that has one hinge.

Tenaculum: A grasping instrument with sharp pointed tips, generally used to manipulate or grasp tissue such as the thyroid or cervix.

Content/methodology/illustrations

Instrument Categorization

Surgical instruments are one of the most important components of a surgical procedure, and they are quite literally an extension of the practitioner's hands. When the surgical team is appropriately acquainted with the name, function and proper handling of the

multitude of instruments available to them, it not only enhances the surgical procedure, but it positively shapes the entire experience for the ultimate end user – the patient.

All instruments are categorized by function. The nine categories are: Accessory; Clamping and Occluding; Cutting and Dissecting; Grasping and Holding; Dilating and Probing; Exposing and Retracting; Aspirating and Suctioning; Stapling and Suturing; and Viewing.

Accessory: An accessory is any instrument that does not fit into a category, but is an important part of the surgical procedure and has a specific function. Examples of accessories are: mallets, harmonic scalpels and guide pins.

Clamping and Occluding: Used to compress or close blood vessels or hollow organs, for hemostasis or occlusion. Examples are: Crile hemostatic forceps and tube occluding clamps.

Cutting and Dissecting: Used to separate or incise, dissect or excise soft or compact tissue with a variety of cutting edges and designs. Examples are: scalpels, Iris scissors and curettes.

Grasping and Holding: Grasping and holding instruments are designed to grip, manipulate and stabilize body tissues. They may also be used to extract foreign bodies. Examples are: Allis tissue forceps, and Backhaus towel clamps.

Dilating and Probing: These instruments are used to probe, expand, examine or create an anatomic passage; to enlarge or open a stricture or introduce another instrument. Examples are: Hank uterine dilator, Sklar mini-dilators/OS finders and probes with eyes.

Exposing and Retracting: Designed to pull aside or hold back wound edges and other tissues to gain access to the surgery or operation site. These instruments are usually called “retractors,” with blunt “blades” that may be curved or angled, or with sharp or dull prongs. Blades vary in size, and some retractors are double ended or come in a variety of shapes, lengths and widths depending on functional needs. Examples are: bone hooks, Volkman retractors and Zalkind retractors.

Aspirating and Suctioning: Aspirating and suctioning instruments are used to remove blood and other fluids and debris from the operating site. These instruments are hollow tipped and come in a variety of sizes and shapes for ease of access. Examples are: cannulas, aspiration suction tubes.

Stapling and Suturing: Used to align, secure and approximate the edges of soft or compact tissue for healing; they may employ implants like sutures, staples, clips wires or screws. Examples are: needle holders, suture needles and terminal end staplers.

Viewing: Viewing instruments allow the practitioner to illuminate a cavity or structure such as a nasal cavity. Examples include: endoscopes, nasal specula and lighted vaginal specula.

Content/Procedures/methods/illustrations

Classification of surgical instrument

Various basic maneuvers are common to all surgical procedures. The surgeon dissects, resects, or alters tissues and/or organs to restore or repair bodily functions or body parts. Bleeding must be controlled during the process. Surgical instruments are designed to provide the tools the surgeon needs for each maneuver. Whether they are small or large, short or long, straight or curved, sharp or blunt, all instruments can be classified by their function. Because the nomenclature is not standardized, the names of specific instruments must be learned in the clinical practice setting. All instruments should be used only for their intended purpose, and they should not be abused.

Dissecting and Cutting Instruments Dissection instruments have sharp edges. They are used to cut, incise, separate, or excise tissues. There are two types of dissecting instruments: sharp and blunt. Sharp dissecting instruments should be kept separate from other instruments, and the sharp edges should be protected during cleaning, sterilizing, and storing. To prevent injury to the handler and damage to the sharp edges, proper precautions are necessary to take during the handling or disposing of all sharps, blades, or scalpels.

Scalpels

The type of scalpel most commonly used has a reusable handle with a disposable blade. Most handles are made of brass; the blades may be made of carbon steel. Blades vary by size and shape (Fig. 19-1); handles vary by width and length (Fig. 19-2). Blades with a numeric prefix of “1” as in a “10” series (e.g., 10, 11, 12, and 15) fit handle size number 3 or 7. Blades with a numeric prefix of “2” as in “20” series (e.g., 20, 22, or 25) fit handle size number 4. Disposable scalpels also are available, but are not weighted the same as metal handles. Most surgeons prefer reusable metal scalpel handles. Blade is attached to a reusable handle by slipping the blade into the grooves on the narrow edge of the handle. An instrument, never the fingers, is used to attach and detach the blade; this instrument, usually a heavy hemostat or Kelly clamp, should not touch the cutting edge. Needle holders are not designed to load scalpel blades and the jaws can become misaligned by excess torque. The following are descriptions of blade and scalpel combinations:

- Number 10 blades are rounded toward the tip and are often used to open the skin.
- Number 11 blades have a linear edge with a sharp tip. Can be used to make the initial skin puncture for tiny deep incisions.
- Number 12 blades have a curved cutting surface like a hook. Commonly used for tonsillectomy.
- Number 15 blades have a short rounded edge for shallow short controlled incisions.
- Number 20 blades are shaped similar to number 10 blades but larger.

An assortment of blades with angulations and configurations for specific uses, such as a Beaver blade, also are used. These blades insert into a special universal handle that secures by turning a screw-in collar.

Figure 1: Diagram of different sizes of surgical blades



Knives

Knives come in various sizes and configurations. Like a kitchen paring knife, they usually have a blade at one end that may have one or two cutting edges. The knives are designed for very specific purposes (e.g., cataract knife). Other types of knives have detachable and replaceable blades (e.g., dermatome). A knife blade may be incorporated into a multifunctional instrument, such as a gastrointestinal anastomosis (GIA) or end-to-end anastomosis (EEA) stapler that cuts and staples tissue simultaneously.

Figure 2: Diagram of a cataract knife



Scissors

The blades of scissors may be straight, angled, or curved, as well as serrated, wedge-shaped, sharp, blunt, or combined sharp-blunt tips.

The handles may be long or short. Some scissors are used only to cut or dissect tissues; others are used to cut other materials. To maintain sharpness of the cutting edges and proper alignment of the blades, scissors should be used only for their intended purpose:

- Tissue/dissecting scissors have sharp or blunt tips. The type and location of tissue to be cut determine which scissors the surgeon will use. Blades needed to cut tough tissues are heavier than those needed to cut fine, delicate structures. Curved or angled blades are needed to reach under or around structures.
- Tissue scissors with blunt tips can be used to bluntly dissect between individual planes before sharply dissecting. Handles to reach deep into body cavities are longer than those needed for superficial tissues.
- Suture scissors have sharp-blunt points to prevent structures close to the suture from being cut. The scrub person may use scissors to cut sutures during preparation if needed.
- Wire scissors have short, heavy serrated blades. Wire scissors are used instead of suture scissors to cut stainless steel sutures. Heavy wire cutters are used to cut bone fixation wires.

Figure 3: Diagram of a dissecting scissor (a) and parts of atypical scissor in (b)



(a)

(b)



Short jaw sharp-tipped scissors are used for deep, confined areas such as the nasal cavity.

Sharp-tipped angled scissors with short jaws are used for vascular surgery.

Dressing/bandage scissors are used to cut drains and dressings and to open items such as plastic packets. The protective tip prevents cutting into concealed structures.

Small scissors with especially wedge-shaped tips such as tenotomy scissors used for blunt and sharp dissection.

Cutters and Saws

Many types of instruments have cutting edges suitable for cutting into or through bone and cartilage. Some have moving parts, such as rib cutters. Others, such as drills, saws, and reamers, are powered by air or electricity and have disposable tips and blades.

Blunt Dissectors

Friable tissues or tissue planes can be separated by blunt dissection. The scalpel handle, the blunt sides of tissue scissors blades, and dissecting sponges may be used for this purpose. Elevators, strippers, and dissectors can be used to remove adherent tissue such as periosteum from bone or dura from the inner aspect of the skull.

Debulking Instruments Debulking instruments include curettes, chisels, osteotomes, gouges, rasps, and files. The purpose of these instruments is to decrease the bulk of firm tissue and not necessarily cut along defined tissue planes.



Figure 4 above showing the diagram of an Osteotome

Biopsy forceps and punches

A small piece of tissue for pathologic examination may be removed with a biopsy forceps

Curettes

Soft tissue or bone is removed by scraping with the sharp edge of the loop, ring, or scoop on the end of a curette.

Snare

A loop of wire may be put around a pedicle to dissect tissue such as a tonsil or a polyp. The wire cuts the pedicle as it retracts into the instrument. The wire is discarded and replaced with a new one after use.

Grasping and Holding Instruments

Tissues should be grasped atraumatically and held in position so the surgeon can perform the desired maneuver, such as dissecting or suturing (approximating), without injuring the surrounding subcutaneous tissues or perforating the skin. Forceps without ring handles are commonly referred to as pick-ups. Delicate Forceps Fine tissues such as eye tissue are held with delicate forceps. They can be toothed or smooth, straight or angled.

Adson Forceps

Forceps are used to pick up or hold soft tissues in approximation during closure. They can be toothed or smooth.

Bayonet Forceps

Forceps are angled like a bayonet to prevent the user's hand from occluding vision in a small space.

Tissue Forceps Also referred to as thumb forceps or pick-ups, smooth forceps resemble tweezers. They are tapered and have serrations (grooves) at the tip. They may be straight or bayonet (angled), short or long, and delicate or heavy. Smooth forceps are atraumatic and will not injure delicate structures.

Toothed Tissue Forceps

Toothed forceps differ from smooth forceps at the tip. Rather than being serrated, they have a single tooth on one side that fits between two teeth on the opposing side or they have a row of multiple teeth at the tip. Heavy types are sometimes referred to as rat-toothed forceps. Toothed forceps provide a firm hold on tough tissues, including skin. Finer versions have delicate teeth for holding more delicate tissue. Care is taken not to perforate the epidermis or dermis when holding tissue.

Allis Forceps

Allis forceps have ringed handles and lock with ratchets. Each jaw curves slightly inward, and there is a row of teeth at the end. The teeth grasp tissue edges securely. Allis forceps can be short or long. Some of the long forceps have a slight curve, such as those used with a hemorrhoid band ligator.

Babcock Forceps

Babcock forceps have ringed handles and lock with ratchets. The end of each jaw of a Babcock forceps is rounded to fit around a tubular structure (i.e., fallopian tube) or to grasp tissue without injury. This rounded section is circumferentially fenestrated. Babcock forceps are straight and can be short or long; they are not occlusive or crushing.

Lahey Forceps: Lahey forceps have ringed handles and lock with ratchets. The jaws of the Lahey forceps have sharp apposing points for grasping tough organs or tumors during excision. They are not occlusive or crushing.

Stone Forceps: Either curved or straight forceps are used to grasp polyps or calculi such as kidney stones or gallstones. Stone forceps have blunt loops or cups at the end of the jaws and do not have ratchets.

Tenaculums

Tenaculums have ringed handles and lock with ratchets and may have a single tooth or multiple teeth, such as a Jacob tenaculum. The curved or angled points on the ends of the jaws tenaculums penetrate tissue to grasp firmly, such as when a uterine tenaculum is attached to the cervix and used to manipulate the uterus during laparoscopy. Some uterine tenaculums have a built-in uterine cannula or probe elevator tip, such as a Hulka tenaculum. A uterine cannula, or probe, can be used during laparoscopy to raise the uterus into the visual field. The cannula tip is inserted into the cervical OS as the tenaculum is clamped on the anterior aspect of the cervix. Dye or contrast media can be instilled through the cannula into the uterine cavity to visualize tubal patency or the inner configuration of the uterine cavity. Some styles have a tenaculum stabilizer tip attached.

Bone Holders Grasping forceps

Vice-Grip pliers, and other types of heavy holding forceps stabilize bone. Some styles have ring handles and locking ratchets. Others have compression grips and do not lock.

Clamping and Occluding Instruments

Instruments that clamp and occlude are used to apply pressure. Some clamps are designed to crush the structure as the instrument is applied and are considered traumatic. The clamped crushed structure is usually sewn, clipped, or electro coagulated and then removed. Other clamps are non-crushing (atraumatic) and are used to occlude or secure tissue, which is restored to patency at some point during the surgical procedure. Hemostatic Clamps Most clamps used for occluding blood vessels have two opposing jaws (with or without serrations or teeth), ringed handles, and lock with ratchets. Most ring-handled clamps have a common design. The length and shape of the shanks or jaws may vary according to the intended function of the instrument.

Hemostats

Hemostats are the most commonly used surgical instruments and are used primarily to clamp blood vessels. They have a crushing action. Hemostats have either straight or curved slender jaws that taper to a fine point. The serrations are longitudinal or horizontal inside the jaws.

Crushing Clamps

Many variations of hemostatic forceps are used to crush tissues or clamp blood vessels. The jaws may be straight, curved, or angled, and the serrations may be horizontal, diagonal, or longitudinal. The tip may be pointed or rounded or have a tooth along the jaw such as on a Heaney clamp or hysterectomy clamps length of the jaws and handles varies.

Many forceps are named for the surgeon who designed the style, such as the Kocher and the Ochsner clamps. Some instruments are designed to be used on specific organs. The features of the instrument will determine its use. Fine tips are needed for small vessels

and structures. Longer and sturdier jaws are needed for larger vessels, dense structures, and thick tissue. Longer shanks are needed to reach structures deep in body cavities. Tissue that has been crushed is usually sutured or stapled.

Noncrushing Clamps

Noncrushing clamps are used to occlude bowel or major blood vessels temporarily, which minimizes tissue trauma. The jaws of these types of clamps have opposing rows of fine serrations, but have a softer hold on tissues. The jaws may be straight, curved, angled, or S-shaped.

Retracting and Exposing Instruments Soft tissues, muscles, and other structures should be pulled aside for exposure of the intended surgical site. **Manual Retractors** Most handheld retractors have an ergonomic handle and blade. The term blade should not be confused with scalpel. The blades vary in width and length to correspond to the size and depth of the incision. The curved or angled blade may be solid for body wall retraction (i.e., Mayo, Richardson, or Kelly retractors), fenestrated like a wire whisk for delicate tissue retraction (i.e., lung retractor), or pronged like a rake for raising tissue edges (i.e., Freeman facelift retractor). The prongs on the blades of a rake can be dull or sharp. Some retractors have different sized blades at both ends with a handle in the middle. Other retractors have traction grooves or serrations for slippery surfaces such as the tongue malleable retractor is a flat length of low-carbon stainless steel, silver, or silver-plated copper that may be bent to the desired angle and depth for retraction. Referred to as ribbon retractors, they can be found in assorted widths from 1/2 inch to 4 inches. Some malleable blades attach into self-retaining or bed mounted retractors. **Hooks** Single, double, or multiple very fine hooks with sharp points are used to retract delicate structures. Hooks are commonly used to retract skin edges during a wide-flap dissection such as a facelift or mastectomy.

Some styles of hooks have ball tips, which cause less trauma to tissues. Some hook styles have elastic attachments that are mounted into a preformed frame over the incision. Care is taken not to perforate the epidermis or dermis when using a hook retractor. Only the subcutaneous tissue is held back.

Retaining Retractors

Holding devices with two or more flat or hooked blades can be inserted to spread the edges of an incision and hold them apart. For example, a rib spreader holds the chest open during a thoracic or cardiac procedure with flat blades. A self-retaining retractor may have shallow or deep blades. Some retractors have ratchets or spring locks to keep the device open; others have wing nuts to secure the blades. Some self-retaining retractors have interchangeable blades of different sizes. Each part of the self-retaining retractor is accounted for at the end of the surgical procedure. Detachable blades or wing nuts can be lost during closure. **Bed-Mounted Retractors** Some self-retaining retractors can be attached to the operating bed for stability when a long surgical incision is planned. A bed-mounted retractor is packaged and sterilized separately in one or two trays because of the numerous blades and attachments. All of the parts are counted items. The component parts include an assortment of blades, connecting ratchets, a mounting frame for the blades, and an upright post that attaches to the side rail of the operating bed at the level of the mattress. The scrub person passes the attaching end of the upright post to the circulating nurse who securely connects the post to the OR bed frame with a screw-bolt. The attached end touched by the circulating nurse is no longer sterile. The upright part of the post remains sterile and the mounting frame is attached by the surgeon at the working height desired. The individual blades and ratchets are placed by the surgeon and the first assistant to attain the exposure needed to perform the surgical procedure. Bed-

mounted retractors are useful in preventing fatigue of the team for long procedures. Care is taken to prevent the upright post from resting against the patient's body. Permanent nerve injury can result from prolonged pressure of an immobile post against a nerve during the surgical procedure.

Needle Holders

Most needle holders have ring handles and locking ratchets. They are used to grasp and hold curved surgical needles. Most needle holders resemble hemostatic forceps; the basic difference is the shortness of the jaws. A needle holder has short, sturdy jaws for grasping a needle without damaging it or the suture material. The jaws are usually straight, but they may be curved or angled. The inner surfaces of the jaws also may differ such as crosshatched, serrated, or have a central notch for stability. The size of the needle holder should match the size of the needle (i.e. heavy jaws for large needles and slim jaws for small needles)

Needle holders can be constructed of various lengths and weights. Heavy snub-nosed needle holders are useful for stainless steel sutures. Long-handled needle holders facilitate needle placement in surgical sites such as the pelvis or chest. Short needle holders are used superficially on the skin or subcuticular tissue. Specialty needle holders may have spring handles with or without locks. The jaws of the micro needle holders used for ophthalmic surgery and microsurgery may be either diamond-cut or crosshatched. They can be curved, angled, or straight. A needle holder should not be placed on a magnetic pad because it may become magnetized and cause needle release problems when suturing. Tungsten Carbide Jaws Tungsten carbide is a hard metal. Jaws with an insert of solid tungsten carbide with diamond-cut precision teeth are designed specifically to eliminate the twisting and turning of the needle in the needle holder.

These diamond-jaw needle holders can be identified by the gold plating on the handles.

Crosshatched Jaws

The serrations on the inside surface of the jaws are crisscrossed rather than grooved. Crosshatching provides a smoother surface and prevents damage to the needle.

Winged Instruments Surgeons can examine the interior of body cavities, hollow organs, or structures with viewing instruments and can perform many procedures through them. Speculums The hinged, blunt blades of a speculum enlarge and hold open a canal (e.g., vagina, rectum or a cavity (e.g., nose. An ear speculum is like a funnel.

Endoscopes

The round or oval sheath of an endoscope is inserted into a body orifice (rigid or flexible) or through a small percutaneous skin incision guided by a trocar assembly (rigid). Each type of endoscope is designed for viewing in a specific anatomic location. Many accessory instruments and robotics are used through endoscopes.

Hollow Endoscopes

In a hollow endoscope, the rigid hollow sheath permits viewing in a forward direction through the endoscope (i.e. bronchoscope, esophagoscope, or sigmoidoscope). The sheath is made of brass, stainless steel, or plastic. A light carrier supplied by a fiberoptic cable provides illumination. A working space is created by manual insufflation of air. Lensed Endoscopes Lensed endoscopes have either rigid or flexible sheaths, and they have an eyepiece with a telescopic lens system for viewing in several directions. Lighting is provided through fiberoptic cable and an electric light source with extremely bright lamps. Used in combination with video-assisted technology, computerization permits recording action videos and still digital photography. Many endoscopes have working

channels for performing specialized procedures. Lensed instruments are complex and require careful handling to avoid damage.

Aspiration, Instillation, and Irrigation Instruments Blood, body fluids, tissue, and irrigating solution may be removed by mechanical suction or manual aspiration. Reusable suction tips aspiration devices have lumens that are difficult to terminally clean and sterilize. Many of these items are available in disposable models.

Aspiration

Aspiration, or suction, involves the application of negative pressure (less than atmospheric pressure) for evacuating blood or fluids, usually for visibility at the surgical site. An appropriate style tip is attached to sterile tubing; many tips are disposable. The style of the suction tip depends on where it is to be used and the surgeon's preference. Disposable styles are preferred because the inner lumen is difficult to clean. Blood, body fluid, or tissue may be aspirated manually to obtain a specimen for laboratory examination or to obtain bone marrow for transplantation. Suction is often performed with a needle and syringe.

Poole Abdominal Tip

The Poole abdominal suction tip is a straight hollow tube with a perforated outer filter shield. It is used during abdominal laparotomy or within any cavity in which copious amounts of fluid or pus are encountered. The outer filter shield prevents the adjacent tissues from being pulled into the suction apparatus. This outer shield is completely removable.

Frazier Tip

The Frazier tip is a right-angle tube with a small diameter. It is used when encountering little or no fluid except capillary bleeding and irrigating fluid, such as in brain, spinal, plastic, or orthopedic procedures. The Frazier tip keeps the field dry without the need for sponging. One model has a connection for an electrosurgical unit, and the tip can be used for fulguration. A fiberoptic cable can be attached to another model. The vacuum exerted by the suction is controlled by a small hole that is covered by the thumb in the handgrip of the Frazier.

Yankauer Tip

The Yankauer tip is a hollow tube that has an angled shaft and a perforated round-ball tip. This suction tip is also referred to as tonsil suction. Large quantities of blood and fluid can be suctioned quickly with a Yankauer tip, which is useful for visualization during ruptured aneurysms. Reusable Yankauer suction tips have a removable end cap that screws on. This must be accounted for at the end of the procedure. It is easily lost in a patient.

Trocar

A trocar assembly may be needed to cut through tissues for access to fluid or a body cavity. A trocar has two parts—a sharp obturator and a sheath or a blunt obturator and sheath. The sharp obturator with the sheath is used to perforate the tissues. When the trocar assembly is in place the obturator is removed, leaving the sheath in position and creating a stented tunnel for drainage of fluids, introduction of instrumentation, or instillation of medication. Some trocars have suction ports, such as is used for draining the gallbladder.

Endoscopic instruments may be manipulated through special trocars that have valves for insufflation, irrigation, and aspiration.

Cannula

A cannula with a blunt end and perforations around the tip may be used to aspirate fluid without cutting into tissue. Cannulae also are used to open blocked vessels or ducts for drainage or to shunt blood flow from the surgical site.

Dilating and Probing Instruments

Malleable copper and brass dilators and probes can be plated with chromium or silver. A set of Hank, Hawkins, Pratt, or Hegar dilators are used to dilate the uterine cervix. A probe is used to explore a path in the structure or to locate an obstruction. Probes are used to explore the depth of a wound or to trace the path of a fistula. Probes and dilation instruments used as tunneling devices can make a passage under the skin for a vascular graft or shunt.

Measuring Instruments Rulers, depth gauges, templates, and trial sizers are used to measure parts of the patient's body.

Instruments many accessories are used in addition to the basic instruments previously discussed. For example, a mallet may be needed to drive a cutting instrument into bone. Screwdrivers are used to affix screws into bone. Each surgical specialty has its own accessories, many of which are described in the specialty chapters. One example is the Hudson brace that is used to manually drill through the cranium

Of the thousands of instruments available, surgeons choose those that are most suitable to meet their particular needs. Differences in the size, curvature, or angulation of jaws or blades; in the length of handles; and in the weight and shape of the instruments can simplify, improve, and even shorten surgical time, which ultimately benefits the patient. Instruments should be maintained to provide the specific function each has been designed to do.

Microinstrumentation

Improved outcomes of surgical intervention using microsurgical techniques are to a great extent the result of the miniaturized precision instrumentation developed in association with the performance of these delicate procedures. The instruments are extremely fine, delicate, and miniature enough to handle in the very small working areas, such as the eye, spine, brain, or other body parts requiring microsurgery. Manipulation becomes more difficult with increased size or bulk and weight. As with techniques, instruments are constantly being improved. Surgeons work with manufacturers to develop appropriate instruments, suture materials, and needles. Many surgeons purchase the instruments of their choice. Whether owned by the surgeon or the facility, microinstruments require exacting care to maintain desired function. Instruments are designed to conform to hand movements under the microscope. They must permit secure grasp, ease of holding and manipulation, and fulfillment of their intended purpose. They are shaped to not obscure the limited field of view. Although these factors are important criteria for any instrument, they are especially vital for microinstruments. Everyone assisting in or setting up for microsurgical procedures must know the identification and functions of these unique instruments. Design is coincident to function. Material and Surface Microinstruments are made of stainless steel or titanium. Titanium alloy is considerably stronger yet lighter weight than stainless steel. Some microinstruments are malleable for desired angling. Some are disposable. All are extremely vulnerable to abuse. Finishes of at least the portions of instruments exposed to light in the surgical field are deliberately dulled during manufacture to reduce glare under the microscope, which is both annoying and tiring to the surgeon. Titanium microinstruments have a dull blue finish. Shape and Tips Microinstruments are shorter than standard instruments and often are angulated for convenience of approach and avoidance of obstruction of the surgical

field. Instrument tips have minimal separation, which is compatible with their function. Finger pressure and movement necessary to close wide tips are undesirable, because they may induce tremor. **Handles**

Handles are designed for a secure and comfortable grasp, with a diameter comparable to that of a pen or pencil. Minimal diameter between fingers facilitates feel and accuracy of manipulation. Double-handled instruments (scissors, needle holders) have a slightly larger diameter than single-handled ones (razor knife). The shape of the handle is also important for manipulation. For example, instruments that are rotated between fingers when in use, such as forceps, must be turned easily. Their handles therefore are rounded or have six sides like a pencil. Those that are not rotated have finger grips or are flattened. Ring-handled instruments are not practical in microsurgery. Many instruments, particularly scissors and some needle holders, have spring handles that return tips to the open position between cutting or grasping functions. The distance from hinge to tip will vary according to function. Proper spring tension can be easily ruined by mishandling. Handles should be long enough for comfort in the working position but must not extend beyond the working distance to contact the unsterile objective of the microscope. The maximum length of most microinstruments is about 4 inches (100 mm). Gripping surfaces should be functionally located to prevent fingers from slipping during manipulation. These surfaces serve as a guide to accurate finger positioning. The gripping area may have a six-sided, round, knurled, or flat and serrated surface. Tactile sense is important to the procedure so the surgeon does not have to look at the instrument to use it.

Self-assessment

1. What is the relationship between the characteristics of a particular type of tissue and the instruments that are used on that tissue?
2. On what types of tissue would you not use an instrument with teeth (one that punctures)?
3. What is the advantage of having a right-angled instrument?
4. What is the most efficient way to learn the names of surgical instruments?
5. Describe a tenaculum.
6. List the correct scalpel blade numbers and their handles.
7. How would you protect the cutting edges of instruments from damage during surgery?
8. What is the function of a self-retaining retractor?
9. Describe your personal plan for learning the names and uses of instruments.

Case study

As a student, you are scrubbed on a large case. Your preceptor has shown you which instruments will be needed during the case. As the preceptor names the instruments, you remember only a few of the names. To make matters more difficult, the preceptor does not tell you the technical name of the instruments but refers to them by the nicknames used at this hospital. What strategies might you use to pass the correct instrument even though you do not know the name?

Reference

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Learning outcome 2: Check and assemble OT instruments and apparatus

This unit trainees will train how to establishing the working condition of Operation Theatre instruments, impact knowledge and skills of referral of faulty Operation Theatre instruments, assemble various parts of Operation Theatre instruments and apparatus conduct Operation Theatre practice and pack and sterilize Operation Theatre instruments

Inspecting surgical instruments

The surgical technologist takes a proactive role in checking instruments for safety and correct operation. This can be done while instruments are being prepared for a procedure, or after the procedure when instruments are sorted and prepared for terminal disinfection. Stainless steel instruments can be damaged mechanically or structurally, or there can be defects on the surface that lead to weakness or cross-infection.

Damaged instruments must be withdrawn from service because they can lead to patient injury, lost operating time, and increased expense.

Scissors

Look for pitting, chipping, and fractures along the blade edges. Vertical cracks in the blade tine are serious and may mean that the instrument cannot be repaired.

Check that the scissor tips meet precisely and are not bent or chipped. Sharp dissection scissors can develop burrs on the tips, usually from misuse or rough handling with heavier

instruments. When the scissors are opened and closed, there should be no grinding of tips or shanks, which would indicate rough edges or bent tines. 3. The center screw of the scissors can become worn with time. When the scissors are opened and closed, they should feel snug but not tight at the center screw. 4. Test the blades for sharpness by cutting through a surgical glove. There should be no resistance, and the blade should cut a straight line. Remove dull scissors from active use as soon as a defect is noticed.

Anesthetic machine or **Boyle's machine** is used by anesthesiologists to support administration of anaesthesia

The most common type of anesthetic machine is designed to provide an accurate and continuous supply of medical gases (such as oxygen and nitrous oxide) mixed with an accurate concentration of anaesthetic vapour (such as isoflurane), and deliver this to the patient at a safe pressure and flow.

Modern machines incorporate a ventilator, suction unit, and patient monitoring devices.

The original concept of Boyle's machine was invented by the British anesthetist Henry Boyle (1875–1941) in 1917.

Components of an Anesthesia machine

A modern anaesthesia machine includes the following components:

Connections to piped hospital oxygen, medical air, and nitrous oxide.

Reserve gas cylinders of oxygen, air, and nitrous oxide attached via a specific yoke with a Bodok seal.

A high-flow oxygen flush which provides pure oxygen at 30-75 litres/minute

Pressure gauges, regulators and 'pop-off' valves, to protect the machine components and patient from high-pressure gases

Flow meters (rota meters) for oxygen, air, and nitrous oxide, low Flow meters for oxygen

nitrous oxide

Updated vaporizers to provide accurate dosage control when using volatile anaesthetics

An integrated ventilator to properly ventilate the patient during administration of anaesthesia

A manual ventilation bag in combination with an Adjustable Pressure Limiting (APL) valve

Systems for monitoring the gases being administered to, and exhaled by the patient

Systems for monitoring the patient's heart rate, ECG, blood pressure and oxygen saturation, in some cases with additional options for monitoring end-tidal carbon dioxide and temperature.

- An anesthesia machine is a device which is used to deliver a precisely known but variable gas mixture including anesthetic and life-sustaining gases to the patient's respiratory system
- Generally, a variable concentration gas mixture of oxygen, nitrous oxide and anesthetic vapor like ether or halothane is obtained from the machine and is made to flow through the breathing circuit to the patient.
- Anesthesia machine is composed of two subsystems:
 - (i) The gas supply-delivery unit, which consists of tubing and flow meters interconnected in parallel; and
 - (ii) The anesthetic vaporizers, which is used to produce an anesthetic vapour from a volatile liquid.

Maintenance of General Anesthesia

The anesthesia provider attempts to maintain the lightest level of anesthesia in the brain compatible with operating conditions. The following are five objectives of general anesthesia:

1. **Oxygenation:** Tissues, especially the brain, must be continuously perfused with oxygenated blood. The color of the blood, amount and kind of bleeding, and pulse oximetry are indicators of the adequacy of oxygenation. Controls on the anesthesia machine and monitors of vital functions keep the anesthesia provider aware of the patient's condition.
 2. **Unconsciousness:** The patient remains asleep and unaware of the environment during the surgical procedure.
 3. **Analgesia:** The patient must be free of pain during the surgical procedure.
 4. **Muscle relaxation:** Muscle relaxation must be constantly assessed to provide necessary amounts of drugs that cause skeletal muscles to relax. Less tissue manipulation is required when muscles are relaxed.
 5. **Control of autonomic reflexes:** Anesthetic agents affect cardiovascular and respiratory systems. Tissue manipulations and systemic reactions to them may be altered by drugs that control the autonomic nervous system.
- Anesthesia Machine. General anesthesia is maintained by inhalation of gases and IV injection of drugs. An anesthesia machine is always used to deliver oxygen-anesthetic mixtures to the patient through a breathing system. The anesthesia machine includes sources of oxygen and gases with flowmeters for measuring and controlling their delivery; devices to volatilize and deliver liquid anesthetics; a gas-driven mechanical ventilator; devices for monitoring the ECG, blood pressure, inspired oxygen, and end-tidal carbon dioxide; and alarm systems to signal apnea or disconnection of the breathing circuit. Breathing tubes of corrugated rubber or plastic carry gases from the machine to the facemask and breathing system. The reservoir (breathing) bag compensates for variations in respiratory demand and permits assisted or controlled ventilation by manual or mechanical compression of the bag. Sterile disposable sets containing tubing, a mask, a Y-connector, and a reservoir bag are commercially available in conductive and nonconductive materials.

Learning outcome 3: Pack and sterilize Operation Theatre instruments

Definition of key terms

Aeration - Warm circulating air is used to remove ethylene oxide sterilant gas from packages in a special chamber for a specified period of time.

Bowie-Dick test- Class 2 chemical indicator used in a prevacuum sterilizer to test the efficacy of the air removal cycle.

Custom packs- Prepackaged collections of disposable supplies, drapes, sponges, and containers prepared by the manufacturer or the distributor according to specific instructions and requests by a particular service at a facility.

Flash sterilization- A rapid method of steam sterilizing properly prepared instruments for immediate use. These instruments are not wrapped. Flash sterilization is not the preferred method of sterilization.

Gravity displacement sterilizer- A steam sterilizer that uses steam in a downward motion to remove air from the sterilizing chamber. Air exits near front lower drain. Can be high speed or pulsing.

Indicator- Devices inserted into a packed set and attached to the outside of the wrapper/container used by health care personnel to monitor sterilization exposure conditions. **Integrator**- A chemical device inserted into a packed set that changes in response the time, temperature, and steam penetration to demonstrate visually that parameters necessary for sterilization have been met.

Prevacuum sterilizer -A faster steam sterilizer that removes air by a vacuum before filling the chamber with steam. Also known as dynamic air removal steam sterilization. Process challenge pack

Sterilization- Processes by which all pathogenic and nonpathogenic microorganisms, including endospores, are killed. This term refers only to a validated process capable of destroying all forms of microbial life, including endospores.

The sterilizer- is a piece of equipment used to attain either physical or chemical sterilization. The agent used must be capable of killing all forms of microorganisms. Strike-through Moisture or other substance penetrates through a sterile drape or wrapper to an unsterile surface. Strike-through provides a pathway for capillary action.

Terminal cleaning- Thorough cleaning and disinfection of the perioperative environment at the end of use.

Contents/methods/procedures/illustrations

Assembly of Instrument Sets

The weight of instruments and density of metal mass are distributed in the tray to allow steam penetration for sterilizing and revaporization for drying. A large tray distributes instruments so they make minimum contact with one another.

The size, design, and density of instruments are more important than their weight. The conditions necessary for steam sterilization are difficult to achieve in exceedingly heavy sets. Trays should not be overloaded. Closed container instrument sets should not weigh more than 25 pounds. Consideration is given to the personnel who must lift and move the packed sets when determining the presterilization weight. To assemble instrument sets for sterilization, the following steps should be performed: 1. Make sure the instruments are thoroughly dry. All instruments belonging to each set must have passed through terminal cleaning and terminal sterilization before they are safe to handle. 2. Unless contraindicated, place an absorbent towel or foam in the bottom of the tray to absorb condensate, as for a rigid container with vacuum valves. Include a biologic indicator or chemical integrator in the tray.

3. Count instruments as they are placed in the tray, and record the number of each type. A preprinted form is often used for this purpose. This form might be placed in the tray before wrapping and sterilization so the circulating nurse and scrub person can verify the baseline count. The form can be folded in half and placed in a paper peel pouch without a plastic coating to prevent printer toner from transferring to the instruments during processing. Concerns for toner particulate transfer to the instruments during the cycle are under study. The best choice may be to affix the tray inventory count sheet to the outside of the package. 4. Arrange the instruments in a definite pattern to protect them from damage and to facilitate their removal for counting and use. Follow the instrument book or other listing of instruments to be included. 5. Place heavy inst 6. Open the hinges and box locks on all hinged instruments. 7. Place ring-handled instruments on stringers or holders designed for this purpose. The curved jaws of hemostatic forceps and clamps should point in the same direction from smallest to largest. Instruments should be grouped by style and classification (e.g., six straight hemostats, six curved hemostats). Do not band with rubber bands. The metal under the band will not sterilize adequately. 8. Place sharp and delicate instruments on top of other instruments. They can be separated with an absorbent material or left in a sterilizing rack with the blades and tips suspended. The blades of scissors, other cutting edges, and delicate tips should not touch other instruments. If the instrument has a protective guard, leave it on. Tip-protecting covers or instrument-protecting plastic sleeves should be made of material that is steam-permeable and does not melt or deform with heat. 9. Place concave or cupped instruments with the cupped surfaces down so that water condensate does not collect in them during steam sterilization and drying.

10. Disassemble all detachable parts. Some parts, such as screws and springs, can be put in a peel pouch that is left open. Sealed pouches may not process correctly during the sterilization process.

11. Separate dissimilar metals. For example, brass knife handles and malleable retractors should be separated from stainless steel instruments. Preferably, put each metal in a separate tray, or separate metals with absorbent material.

12. Place instruments with a lumen, such as a suction tip, in as near a horizontal position as possible. These instruments should be tilted as little as possible to prevent trapped air or the pooling of water condensate.

13. Distribute weight as evenly as possible in the tray. Some trays have dividers, clips, and pins that attach to the bottom, which help prevent instruments from shifting and keep them in alignment.

14. Wrap the tray, or place it in a rigid container. Check woven textile wrappers for holes or abrasion. Sequentially double wrap in a woven or nonwoven material, or use a double thickness wrap in a single-fold configuration.

15. Place a chemical indicator tab or tape on the outside wrapper or container for proof that the package has been through the parameters necessary for sterilization.

16. Label the sets appropriately with their intended use (e.g., basic set), the date sterilized, and the process lot control number.

Packaging Instruments and Other Items for Sterilization To be effective, the sterilizing agent must come into direct contact with all surfaces of every instrument. Therefore instruments must be packaged (individually or in sets) in such a way to allow adequate exposure to the sterilant, prevent air from being trapped and moisture from being retained during the sterilization process, and ensure sterile transfer to the sterile field.⁴ The majority of surgical instruments are made of stainless steel and can be sterilized by steam under pressure. Effective steam sterilization involves the direct contact of all surfaces with steam and the revaporization of water condensate to produce a dry, sterile instrument.

Instrument Packaging For sterilizing and transporting, instruments are put in a closed container or wrapped individually. Instruments placed in open trays are wrapped as sets. Instruments may be sterilized unwrapped in a high-speed steam sterilizer immediately before use (not recommended), may be prepared in advance as for a case cart, or may be retained in sterile core storage until needed.⁴ **Packaging Considerations** The packaging materials for all methods of sterilization should do the following:⁴

- Permit penetration of the sterilizing agent to achieve sterilization of all items in the package.
- Allow the release of the sterilizing agent at the end of the exposure period and allow adequate drying or aerating.
- Withstand the physical conditions of the sterilizing process.
- Maintain integrity of the package at varying atmospheric and humidity levels. In dry climates or at high altitudes, some packaging materials are susceptible to rupture during sterilization or dry out and crack in storage.

- Provide an impermeable barrier to microorganisms, dust particles, and moisture after sterilization. Items must remain sterile from the time they are removed from the sterilizer until they are used.
- Cover items completely and easily and fasten securely with adhesive strip or a heat seal that cannot be resealed after opening. Seal integrity should be tamperproof. A margin of at least 1 inch (2.5 cm) is considered a standard for safety on all sealed packages. Scissors should not be used to cut off the end of a package. Contents cannot be drawn out over this cut end because the contents would be contaminated by the edge of the packaging material. For the same reason, packages are never torn open below a seal.
- Resist tears and punctures in handling. If accidental tears and holes do occur, they must be visible.
- Identification of the contents and the chemical indicator evidence of exposure to a sterilizing agent should be placed on a label that is visible on the outside of the package.
- Be free of toxic ingredients and nonfast dyes.
- Be lint-free or low-linting.
- Protect the contents from physical damage.
- Permit easy removal of the contents with transfer to the sterile field without contamination or delamination (separation into layers).
- Be economical. Packages should be wrapped far enough away from sterile storage areas so that mixing sterile and nonsterile packages is not possible. Cabinets that contain nonsterile items should be labeled conspicuously. A procedure for sending items to and receiving items from the sterilizer should be set up so that sterile and nonsterile packages can never be confused en route. This procedure must be understood by everyone. Packages may be wrapped with nonwoven or woven materials sequentially in two layers. Double-thickness wrapping in one layer without using the sequential method is also acceptable. The wrappers can be stitched around the edges or bonded/fused together. Aseptic presentation to the sterile field is the prime consideration. Each facility should determine which method and material is best suited to the clinical environment.

Handling Powered Instruments Before any new or repaired powered instrument is put into use, the biomedical technician should verify that the instrument is functioning according to the manufacturer's specifications. Powered instruments are not without some inherent dangers. Key points in handling powered instruments include the following:

1. Set the instrument and attachments alone on a small sterile table with the safety switch on when they are not in use. This provides added protection from inadvertent activation.
2. Handle and store the air hose or electrical power cord with care. A broken air hose can whip out of control. A broken electrical cord can short-circuit the instrument. Always inspect the hose and cord for cracks and breaks.

3. To prevent inadvertent activation, assemble the appropriate handpiece, attachments (e.g., blades, drills, reamers), and power source with the safety mechanism in position. Always be certain that attachments are completely seated and locked in the handpiece.

4. Test whether the instrument is in working condition before the surgeon is ready to use it and before it is applied to the patient. To prevent inadvertent activation, the safety mechanism must be set in position until ready to use and when changing attachments.

Cleaning and Sterilizing Powered Instruments Powered instruments should always be operated, cleaned, and sterilized according to the manufacturer's directions for use and care. Each instrument has different cleaning, lubricating, packaging, and sterilizing requirements because of its various component parts. The bioburden is determined by the size, design, complexity, and condition of the instrument; the degree of contamination during use; and subsequent decontaminating and cleaning procedures. Microorganisms can become entrapped around the seals on rotating shafts. For instruments with a specific biologic challenge, manufacturers test in the area most difficult to sterilize and recommend sterilization cycles accordingly. The following general guidelines apply to the care of all powered instruments:

1. Clean and decontaminate the instrument immediately after use to maintain optimal function.

a. During the surgical procedure, the scrub person should wipe off any organic debris between uses.

b. The accessories are disassembled for cleaning. Do not process the powered instrument with a bit, burr, or blade in place. The metal against metal prevents sterilization of the unit. If the blade, burr, or bit needs to be changed during the procedure the exposed unsterile surfaces will contaminate the field.

c. The air hose should remain attached to the handpiece during cleaning. The air hose and electrical cord should be wiped with detergent, damp cloth, and dry towel.

d. The motor is not immersed in liquid. The power mechanism cannot be cleaned in a basin of solution or put in a washer-sterilizer, a washer-decontaminator, or an ultrasonic cleaner. The surface of the instrument is wiped with a mild detergent, and caution is used to prevent solution from entering the internal mechanism. The detergent is wiped off with a damp cloth, and the mechanism is dried with a lint-free towel.

2. Lubricate the instrument as recommended by the manufacturer.

Handling Instruments Surgical instruments are expensive and represent a major investment. Surgical procedures have become more complicated and intricate, and as a result, instruments have become more complex, more precise in design, and more delicate in structure. Abuse, misuse, inadequate cleaning or processing, or rough handling can damage and reduce the life expectancy of even the most durable instrument, and the cost of repair or replacement becomes unnecessarily high. Instruments do deteriorate from normal use, but with proper care an instrument should have a life of 10 years or more.^{2,5}

Setting Up the Instrument Table Standardized basic sets of sterile instruments are selected for each specific surgical procedure. A set is a group of instruments that may include all appropriate classifications of instruments or the instruments needed for a specific part of the procedure (e.g., a gallbladder set). The surgeon may prefer some specific instruments that are wrapped separately or added to the instrument set. Instruments are usually prepared, wrapped, and sterilized several hours or days before the surgical procedure so they are dry and cool for safe handling.¹ Instruments are sometimes steam sterilized

immediately before use if there is no other alternative, but this method is not recommended as a routine practice.³ Key elements of decontamination and processing might be skipped, leaving the instruments unready for sterilization. The scrub person counts all instruments, sponges, and sharps with the circulating nurse. Some sets will have printed count sheets enclosed. Observe for incidental transfer of toner from the count sheet to the instruments during steam sterilization. Instruments with fused-on toner should not be used in patient care. Essential points in handling instruments before the surgical procedure include:

1. Handle loose instruments separately to prevent interlocking or crushing. a. Instruments are never piled one on top of another on an instrument table; they are laid side by side on a rolled sterile towel. Close the box locks on the instrument to avoid entanglement when placed on the Mayo stand. b. Microsurgical, ophthalmic, and other delicate instruments are vulnerable to damage through rough handling. Tip protectors are removed and discarded during setup. c. Metal-to-metal contact should be avoided or minimized. Scalpel blades should not be set in a metal basin. The edges can become dull, and small chips of metal can become dislodged and inadvertently transferred to the patient during irrigation.

2. Inspect instruments such as scissors and forceps for alignment, imperfections, cleanliness, and working condition. Unclean instruments render the entire set contaminated.^{1,4} Remove any malfunctioning instrument from the set. It should be labeled and sent for repair. Replace it in the set with a correctly functioning instrument before processing. If the procedure is in process, the circulating nurse can obtain a sterile replacement and remove the broken instrument from the room. Be sure to reconcile the count sheet. a. Scalpel blades should be properly set in handles using a heavy instrument, not fingers. b. Teeth and serrations should align exactly. c. Tips should be straight and in alignment. d. Scissors should be snug and sharp in action. e. Cannulae should be clear and without obstruction. Stylets should be removed.

3. Sort instruments neatly by classifications.

4. Keep ring-handled instruments together with the curvatures and angles pointed in the same direction positioned smallest to largest. Do not permit the instruments to become tangled on the towel roll. a. Hang ring handles over a rolled towel or over the edge of the instrument tray or container within the sterile field. b. Remove instrument stringers or holders if used to keep box locks open during processing. c. Close box locks on the first ratchet as the setup progresses.

5. Leave retractors and other heavy instruments in a tray or container, or lay them out on a flat surface of the table. Do not permit any instrument to hang over the edge of the sterile field.

6. Protect sharp blades, edges, and tips. They should not touch anything. Take care not to perforate the sterile table cover. a. Sets of instruments, such as osteotomes or microsurgical instruments, may be in sterilization racks so that the blades and tips are suspended. These instruments can remain in racks during the initial table setup and until they are needed during the surgical procedure. b. Tip-protecting covers or instrument-protecting plastic sleeves should be removed and discarded before the instruments are used on the patient.

Most covers and sleeves are not radiopaque and could become a retained foreign body. c. If they are not in a rack, handles should be supported on a rolled towel or gauze sponge. This keeps the blades and tips of microinstruments suspended in midair for easy visualization. Handling Instruments During the Surgical Procedure Efficient instrument handling throughout the surgical procedure is the hallmark of an efficient scrub person.⁶ Key points in handling instruments during the surgical procedure include the following:

1. Know the name and appropriate use of each instrument. Using fine instruments for heavy tissue damages the instrument.

2. Handle instruments individually. Tangled instruments are hard to separate in an emergency.⁶ Instruments on the Mayo stand should be closed to minimize fumbling for the correct item. a. If several instruments of the same type will be needed in rapid succession (e.g., hemostats to clamp subcutaneous vessels), three or four may be picked up at one time, but they are closed and passed individually to the surgeon and/or first assistant. b. Instruments with sharp edges and fine tips are more susceptible to damage than are standard instruments. The edges are easily dulled, and the tips are easily bent or broken. Extreme caution is necessary to prevent catching the tips of microinstruments on any object that could bend them.

3. Hand the surgeon or assistant the correct instrument for each particular task. Remember the following principle: Use for the intended purpose only. a. Avoid placing fingers in the instrument rings as the instrument is passed. The instrument may inadvertently drop or snag on drapes, causing an untoward injury to the patient or a team member. The instrument may fall to the floor, thus becoming damaged and contaminated. b. Many surgeons use hand signals to indicate the type of instrument needed. An understanding of what is taking place at the surgical site makes these signals meaningful. c. Select instruments appropriate to the location of the surgical site; short instruments are used for superficial work, and long ones are used for work deep in a body cavity. Experience will facilitate instrument selection according to the surgeon's preference and need. d. Many instruments are used in pairs or in sequence.⁶ When the surgeon clamps and/or cuts tissue, he or she will usually request suture. To use suture, the surgeon will need to grasp the tissue with a pick-up for stabilization. After using suture, the surgeon or assistant will need scissors to cut or a hemostat to hold the end of the strand as a tag. e. Hand instruments around the incisional area, not directly over it, to prevent possible injury. f. Knowledge of anatomy is useful for determining which instrument is needed.

4. Pass instruments decisively and firmly. When the surgeon extends his or her hand, the instrument should be slapped or placed firmly into his or her palm in the proper position for use. In general, when passing a curved instrument, the curve of the instrument aligns with the direction of the curve of the

surgeon's hand. The following points should be remembered when passing an instrument to the surgeon: a. If the surgeon is on the opposite side of the operating bed, pass across right hand to right hand (or with the left hand to a left-handed surgeon). The curve of the instrument should match the curve of the surgeon's hand. b. If the surgeon or assistant is on the same side of the operating bed and to the right, pass with your left hand; if the surgeon or assistant is to your left, pass with your right hand. c. Ring-handled instruments

are held near the box lock by the scrub person and passed by rotating the wrist clockwise to place the handle directly into the surgeon's waiting hand (Fig. 19-46). d. Clip applicators are held between the fingers by the hinged joint during loading and passing. Placing fingers in the rings may cause the clip to be discharged unintentionally. The loaded applicator is passed so the rings automatically pass over the surgeon's finger in a position of function for rapid use. e. Sharp and delicate instruments may be placed on a flat surface for the surgeon to pick up (e.g., "no touch zone").⁶ This technique avoids potential contact with items such as cutting blades, sharp points, and needles in hand-to-hand transfer. Always protect the hands when manipulating sharp instruments. Some surgeons prefer to have all instruments passed directly so great care is taken to place the instrument in the waiting hand without causing the surgeon to look up away from the field. This is common in microsurgery.

5. Watch the sterile field for loose instruments. After use, wipe the blood from the surface and place them promptly on the Mayo stand or instrument table in a position of function.¹ The weight of instruments can injure the patient or cause postoperative discomfort. Keeping instruments off the field also decreases the possibility of their falling to the floor.

6. With a moist sponge, wipe blood and organic debris from instruments promptly after each use. This is referred to as point of use cleaning.¹ a. Demineralized, sterile distilled water should be used to wipe instruments. Saline, blood, and other solutions can damage surfaces, causing corrosion and, ultimately, pitting. b. Blood and debris that are allowed to dry on surfaces, in box locks, and in crevices increase the bioburden that could be carried into the surgical site. Dried material is harder to remove during processing.⁴ c. A nonfibrous sponge should be used to wipe off microsurgical, ophthalmic, and other delicate tips. This type of sponge prevents the snagging and breaking of delicate tips, and the

• FIG. 19-46 Passing an instrument. Tip is visible; hand is free. Handle is placed directly into waiting hand. Avoid potential for lint is decreased. Commercial microsurgical instrument wipes are available.

7. Flush the suction tip and tubing with sterile distilled water periodically to keep the lumens patent.¹ Use only a few milliliters of solution if using irrigating fluids from the surgical field. Keep a tally of the amount used to clear the suction line, and deduct this amount from the total used to irrigate the surgical site. Accurate accounting of the solutions used for patient irrigation is necessary when determining the amount of blood lost during the surgical procedure.

8. Remove debris from electrosurgical (ESU) tips to ensure electrical contact. Disposable abrasive tip cleaners are helpful for maintaining the conductivity and effectiveness of the surface of the tip. Do not use a tip cleaner on Teflon-coated ESU tips. Avoid using a scalpel blade to clean electrosurgical tips, because the debris may become airborne and contaminate the surgical field.

9. Place used instruments that will not be needed again (except sharp, cutting, delicate, or powered instruments) into a tray or basin during or at the end of the surgical procedure. a. Blood and gross debris are removed at the point of use before actual cleaning and

decontamination in the processing area.1 b. Carelessly dropping, tossing, or throwing instruments into a basin causes damage. c. Instruments that have been wiped can be immersed in a basin of sterile demineralized distilled water, not saline solution. The sodium chloride in saline solution and in blood is corrosive and can damage instrument surfaces. Bloody instruments should not soak in a basin of solution for a prolonged period. d. Heavy instruments such as retractors should not be placed on top of tissue and hemostatic forceps and other clamps. Place them in a separate tray. e. Reusable sharps should be kept separate from other instruments of the same or similar size to prevent injury to instrument processing personnel. f. Keep instruments accessible for final counts.

Dismantling the Instrument Table

The scrub person breaks down the instrument table at the end of the surgical procedure while wearing the gown and full PPE. This process may be delayed and the scrub person and instrument table maintained as sterile for a short time if the patient is in critical condition. The breakdown should not be started until a critical patient is out of the OR and admitted to PACU. Some examples of when to wait before breakdown include cardiac, vascular, some neuro, multiple trauma, or other procedure or patient condition that could precipitate expedient return to OR for immediate surgical reopening. When the scrub person is assured that the patient will not return to the OR, the table breakdown can begin. Whether used or unused, all instruments on the instrument table are considered contaminated and must be promptly and properly decontaminated/cleaned, inspected, terminally sterilized, and prepared for subsequent use. Wearing gloves, a gown, a mask, and protective eyewear, the scrub person prepares instruments for the cleaning process. Instruments are cleaned in a designated instrument processing area, not in the OR. Key points in handling instruments when dismantling the instrument table include the following:

Check drapes, towels, and table covers to be sure that instruments do not go to the laundry or into the trash. A final quick count is a safeguard. 2. Collect instruments from the Mayo stand and any other small tables, and collect those that may have been dropped or passed off the sterile field. 3. Separate delicate, small instruments and those with sharp or semisharp edges for special handling. 4. Disassemble all instruments with removable parts to expose all surfaces for cleaning. 5. Open all hinged instruments to expose box locks and serrations. 6. Separate instruments of dissimilar metals. Instruments of each type of metal should be cleaned separately to prevent electrolytic deposition of other metals. 7. Flush cold, distilled water through hollow instruments or channels, such as suction tips or endoscopes, to prevent organic debris from drying. 8. Rinse off blood and debris with demineralized distilled water or an enzymatic detergent solution. 9. Follow the procedures for preparing each instrument for decontamination or terminal sterilization. Procedures vary depending on the type of instrument and its components and on the equipment available and its location. a. Some air-powered instruments can be lubricated with a sterile lubricant after sterilization and just before use. b. Some manufacturers supply lubricant (usually a silicone based oil). c. Some instruments must be run after lubrication to disperse the lubricant through the mechanism.

10. Wrap the instrument for sterilization. a. Most manufacturers supply sterilizing cases for powered equipment. These cases can be wrapped in woven or nonwoven material. b.

The instrument must be disassembled. c. The sharp edges of accessories must be protected.
d. Hoses or cords should be loosely coiled.

- **Packing techniques**

Packaging of Sterile Products

Contents

- Why packing of sterile goods?
- Packaging concept for sterile products
- Primary packaging containing the product
- Secondary packaging
- Transport packaging
- Validation of the packaging system
- Requirements for the primary packaging
- General guidelines for packaging practice
- Packaging materials
- Soft/flexible packaging materials
- Textile sheets: cotton or linen
- Paper sheets
- Paper sterilization bags
- Non-woven sheets
- Laminated film pouches
- Rigid packaging systems
- Sterilizing drums
- Sterilizing containers
- Wrapping techniques for packaging using sheets
- Accessories for packaging
- Indicator tape

- Trays and baskets
- Protection materials
- Dust covers
- Workstation for packaging
- Choice of materials and methods for packaging of sterile supplies (primary packaging)
- Textile packs
- Small quantities of textile and/or bandages/swabs
- Instrument sets in trays/baskets
- Individual instruments:
- Bowls and trays (small)

Bowls and trays (large)

Catheters, tubing, hoses

Scopes

Fine surgical instruments (individual as well as in sets)

Mammal prosthesis

Normative references (European)





Why packing of sterile goods?



Packaging is essential for maintaining sterility!

At the end of a correct sterilization process, products inside the sterilizer chamber are sterile. The air in the room where the sterilizer is installed contains dust particles, which may carry micro-organisms. So when taking out the load from the sterilizer, it soon will be contaminated again. In addition usually sterile goods are stored for quite some time before they are used, and moreover they are transported through the hospital to the place they are to be used. It thus is obvious, that, when not protected, the goods will definitely be recontaminated by the time they are used.

The goods must be put in a packaging to prevent recontamination after sterilization. At the same time the packaging should be suitable for sterilizing the goods it contains!

Moreover the packaging protects its content against damage during handling and transport.

Packaging concept for sterile products

Depending on use, storage and transportation, a sterile product should be packaged in one or more packaging layers:

Primary packaging containing the product



An instrument set in its primary packaging

The primary packaging prevents recontamination of the product after sterilization. It should provide an effective microbial barrier and it should allow passage of air and the sterilizing agent, e.g. steam. A primary packaging is sufficient in situations that there is no chance that dust is deposited on the pack. In a dust free storage or in case of immediate use of the materials. The primary packaging maintains sterility during storage and transport. Examples of primary packaging: 2 layers of paper, 2 layers of non-woven sheets, single or double laminated film pouch, paper bag or container with adequate filter(s).

Secondary packaging



Disposable sterile items in secondary packaging

This second layer used to facilitate proper storage and internal transport to user. It may contain one or more primary packages. E.g. an extra, plastic wrapper, bag, cardboard box or container. It may offer extra protection against dust and gives extra mechanical protection, making handling easier. Commonly used for disposable sterile supplies.

Transport packaging



Closed trolleys for external transport

A transport package is used for external transporting of sterile goods in their primary and secondary packaging. Usually it is a strong cardboard box, cradle, closed trolley or other type of container.

When the goods enter the clean area, e.g. operating theatre, the transport packaging should be removed.

Validation of the packaging system



Any combination of load, packaging and sterilization process should be validated.

It is essential that a packaging system with its content meet the requirements in terms of sterility maintenance and protection of its contents. That is why any packaging should be validated in combination with the actual load and the sterilization process used. When testing a packaging system, also handling transport and storage conditions should be considered! (see EN 868-1)

Requirements for the primary packaging

When observing a packaging from the moment it is put around a product before sterilization until the moment it is opened, just before use of its content, the requirements become clear:

Enabling sterilization. The packaging should allow air that is in the pack to be driven out and the sterilizing agent to reach all surfaces of its content.

Compatible with the sterilization process. The packaging should be able to withstand the conditions during the sterilization process such as pressure changes, high temperature and humidity

Maintaining Sterility. After taking the product out of the sterilizer it should remain sterile during handling, transport and storage until use.

Strong. It should be remain intact after any intended handling and shipping

Ensure product integrity and patient safety. It should not release any chemicals or particles or affect the product in any other way, which may affect the quality of the product or which might endanger the patient on which the product will be used.

Indicator. The packaging should bear a clearly visible marking indicating whether or not the product is sterilized.

Facilitate aseptic opening and presentation. When opening a packed sterile product, the chance for recontamination of the product should be kept as low as possible. When taking the product from the packaging, recontamination should be prevented. In other words: the packaging should facilitate aseptic opening and presentation. This implies:

simple opening

when taking out sterile objects form the packaging, it should not be possible to touch the unsterile outer side of the packaging.

Visible that it was opened. Opening should result in a clearly visible indication that the pack was opened.

General guidelines for packaging practice

If in a health facility more than one sterilization methods are used, the accidental exchange of packaging materials shall be prevented. The packaging should have an indication showing clearly for which sterilization method it should be used.

Reusable medical supplies that are to be re-sterilized (e.g. due to passing of the expiry date of a pack, or because of damaged packaging), should inspected, where necessary cleaned and repacked. Textile should be reconditioned.

The content of a pack, which was opened by mistake, should be considered as non-sterile. It should be made clearly visible that a package was opened. E.g. by clearly damaging the packaging material. The content of such pack shall be inspected and when necessary, cleaned. It then is repacked and sterilized.

Packaging materials

A wide range of materials used for packaging of sterile supplies are available.

Traditionally, packaging materials for sterile supplies were reusable, such as sterilizing drums and cotton ware. Due to their inadequate microbial barrier, most of these traditional materials do not meet the requirements for primary sterile packaging anymore. They may still play a role as mechanical protection or additional dust protection layer. At the moment non-wovens, laminated film pouches, paper bags and containers are used as primary packaging materials. The following is an overview of packaging materials in use in sterile supply:

Soft/flexible packaging materials

Textile sheets: cotton or linen

Use: Inner wrapping of instrument sets or outer dust protection

Textile alone is not suitable as primary packaging!

Cotton sheets have long been the standard packaging material for sterile goods. It has some major advantages

Textile always has been a very common and well known hospital commodity

Strong

Well drapable and convenient in use

Can be reused.



Packaging in cotton still being used in health facilities with limited resources

The openings between the threads however are larger than most micro-organisms and thus the fabric does not provide an adequate microbial barrier. It therefore does not meet the requirements anymore as primary packaging for sterile goods. They however are still often used as an inner wrapping for protection or as an outer dust cover.

Whenever textile is used, it should contain its natural humidity. (It should be conditioned). If textile is too dry it may cause overheating of the steam and thus cause a

failing

sterilization.

Paper sheets



Paper sheets available in many sizes and strengths

Use: Primary packaging for wrapping of textile packs and instrument sets in trays. Also used as inner packaging in containers

Paper was the first alternative that replaced textile. It has a smaller pore size than textile. And thus can be used as primary packaging. Smooth papers are used for inner packaging, whereas crepe paper is stronger and is rough. Crepe paper can be used for inner and outer packaging.

During sterilization, steam penetrates through the packaging. When paper is wet, it loses much of its original strength. Therefore stress in paper should be prevented. Wrapping should not be too tight, but also not too loose. It is essential that drying is adequate.

Paper sheets are for single use only.

Paper sterilization bags

Use: For packaging of individual instruments or small sets used in nursing stations and wards.

Closing is usually done in a sealing device.

Disadvantages:

They are not very strong

Opening is not convenient: tearing or cutting

They do not facilitate aseptic opening.

You cannot see what is inside



Paper sterilization bags

Aseptic presentation can be improved by putting instruments in the bag with the handle at the opening end. It is not convenient to take out instruments from the bag. Its use has decreased with the introduction of laminated film pouches.

Paper sterilization bags are for single use only

Non-woven sheets

Use: Primary packaging for wrapping of textile packs and instrument sets in trays. Also used as inner packaging in containers



Non-woven wrapper



Non-wovens are water-repellent



Packaging an instrument set in a non-woven sheet using the parcel fold

Non-wovens contain a certain amount of synthetic fibers. Added may be inorganic, textile, cellulose or other kind of synthetic fibers. These fibers of different materials are joined together, by for example pressing and heating. This means that the fibers are not woven together, but sealed together. For sterilization special non-wovens have been developed to meet the requirements for the primary packaging of sterile goods. They combine the good characteristics of other packaging materials:

Very strong

Well drapable

Allow air removal and penetration of the sterilizing agent

Very small pores, thus efficient microbial barrier

Virtually lint-free; free of particles and loose fibers

They repel liquids (hydrophobic) Fluids are not absorbed into the fabric.

Various non-woven materials are available for a range of applications in the sterilization department: extra softness, extra strong, etc.

Non-woven sheets are for single-use only

Laminated film pouches

Use: Primary packaging for individual instruments or small instrument sets.



Typical laminated film pouch

They were the follow up of paper sterilization bags. The pouches consist of a sheet of paper or non-woven and a sheet of laminated transparent plastic, which are sealed together. The film cannot be penetrated by steam or air. Removal of air and penetration of steam is through the paper/non-woven. The pouch is opened by peeling back the laminated sheet from the paper sheets, like opening a banana.



Laminated film pouches in various sizes and on roll



Sealing device for pouches

Pouches are available in many sizes. The open end of the pouch is closed with a sealing device. It is essential that the sealing temperature and pressure are adjusted well in order to get an adequate seal.

Also laminated film packaging is available on roll. The user can cut pouches to any size needed. In that case both sides need to be sealed by the user.

Remarks

The peel-open system assures dust-free aseptic opening and presentation.

The pouch should be such that when peeling it open, not the paper nor the laminate will tear. It should open neatly along the seals.

It should not release fibers or fluffs.

The content is clearly visible.

It should not be possible to reseal the pouch, when opening by mistake.

A sterilizer indicator should be on the pouch indicating whether a product was processed.

The content should not be tightly surrounded by the packaging material; it should be able to move freely inside the pouch.

Sterilization pouches should be put upright in a grid basket or container and not too tight together, such that a hand can slide in between them.

Laminated film pouches are for single use only.

When packaging in dual laminated pouches, make sure that the paper of both packages are on the same side. The inner pouch should fit freely in the outer pouch!

Rigid packaging systems

Sterilizing drums

Use: As secondary packaging layer: mechanical protection of its content

Not suitable as primary packaging.

In the past, most sterilizers had vertical, round chambers. For various chamber sizes, drums were available. To allow steam into the drum, the wall is perforated. A metal band around the drum can be shifted in order to open and close the perforations. Before sterilization they are opened. After sterilization the perforations are closed again. They are known as Schimmelbush drums and were widespread.



Sterilizing drums are only suitable as additional mechanical protection (secondary packaging)

Problems:

The band is closed after sterilization and thus the content may be recontaminated immediately after sterilization.

The band does not provide an adequate closure, thus contaminated air may be sucked inside the drum. Also the lid closure does not provide a proper seal.

Most modern autoclaves have horizontal, rectangular chambers. Round drums are then very inefficient in using the sterilizer space.

The drums have hinged lids, hindering aseptic opening.

Therefore they are not suitable as primary packaging. They may only be used as an extra mechanical protection.

Drums with filters in top and bottom

Drums are available with filters in the lid and bottom. They can be considered as sterilizing containers for round, vertical autoclaves.

Sterilizing containers

Use: Primary packaging of textile packs and instrument sets in trays.

Well designed sterilizing containers offer the following features:

Air removal and steam access through filters or valves. They ensure maintenance of sterility during storage.

For autoclaves with downward displacement processes containers are available with filters in the lid and bottom.

The sizing of most container systems is based on the Sterilizing Unit (1 StU = 30cmx30cmx60cm). Thus they fit in sterilizers, instrument washer disinfectors, and are compatible with packaging materials, that are also based on this unit.

The lid is removed completely from the bottom part thus facilitating aseptic opening

They provide an effective mechanical protection during handling and transport

They can be stacked. However stacking should not hinder steam penetration, nor cause entering of condensate dripping down from higher containers. Proper drying is to be confirmed.

Grid baskets can be stacked inside containers.

Labeling system offers clear identification of the content, production data etc.

A tamper protection device shall give clear indication whether a container was opened

Depending on the sterilizer process, containers shall be used with filters in top and bottom (for downward displacement process) or containers with filters only in the top (fractionated prevacuum process).



Loading a sterilizer with containers



Sterilizing container



Taking out a tray from container

Wrapping techniques for packaging using sheets

When opening a pack containing sterile materials, it is essential that, due to the act of opening, the content is not contaminated. Wrapping techniques for packs and sets have been developed which assures this aseptic opening. The most common wrapping techniques that are applied for packaging of textile packs and instrument sets are the envelope fold and the parcel fold. The unfolded wrapper covers the instrument table and thus provides a sterile field. The techniques can be used for sheets of textile, paper and non-wovens.

11. Sterilize the instrument in steam unless contraindicated by the manufacturer.

Types of Sterilization methods

Dry Heat Sterilization

Dry heat in the form of hot air is used primarily to sterilize anhydrous oils, petroleum products, and talc, which steam and ethylene oxide gas cannot penetrate. The destruction of microbial life by dry heat is a physical oxidation or slow burning process that involves coagulating the protein in cells. Higher temperatures are required in the absence of moisture because the microorganisms are destroyed through a very slow process of heat absorption by conduction. Dry heat sterilizers are not commonly found in the OR processing areas. Dry heat is commonly used in dental suites. Advantages of Dry Heat Sterilization

Hot air penetrates certain substances that cannot be sterilized by steam sterilization or another method.

Dry heat is a protective method of sterilizing some delicate, sharp, or cutting-edge instruments. Steam may erode or corrode cutting edges. Disadvantages of Dry Heat Sterilization

A long exposure period is required, because hot air penetrates slowly and possibly unevenly.

The time and temperature required will vary for different substances.

Overexposure may ruin some substances.

Types of Dry Heat Sterilizers

Mechanical Convection Oven. The most efficient and reliable dry heat sterilizer is an electrically heated, mechanical convection hot air oven. A blower forces hot air in motion around items in the load to hasten the heating of substances and to ensure a uniform temperature in all areas of the oven. Early models operated at 320° F to 340° F (160° C-171° C) for 1 to 2 hours. Faster portable tabletop models are available; these run at 375° F to 400° F (190.5° C-204° C), with total cycle times of 6 minutes for unwrapped items and 12 minutes for wrapped ones. Optional cooling chambers are also available. **Gravity Convection Oven.**

A conventional gravity displacement steam sterilizer chamber can be used for dry heat sterilization. Heat is provided by steam in the jacket only; this heat may not be evenly distributed throughout the chamber. The hot air rises initially and by gravity displaces cooler air at the bottom of the chamber. The maximum temperature that can be obtained is 250° F (121° C) or 270° F (132° C) in a high-pressure gravity displacement sterilizer. To ensure adequate heat conduction through all items, the exposure period is a minimum of 6 hours and preferably overnight.

Loading the Sterilizer When loading a dry heat sterilizer, it is necessary to allow space between the items and along the chamber walls so the hot air can circulate freely. The chamber is never loaded to full capacity. **Timing the Load.** The time of exposure in a dry heat sterilizer varies depending on the characteristics of individual items, the layer depth in containers, and the temperature in the sterilizer. The timing of exposure begins when the thermometer registers the desired temperature. Items are exposed for the following minimum periods, assuming the amount in each container is kept to a minimum and the sterilizer is loaded according to the manufacturer's recommendations:

6 minutes at 400° F (204° C), unwrapped

12 minutes at 375° F (190.5° C), wrapped

1 hour at 340° F (171° C), wrapped or unwrapped

2 hours at 320° F (160° C), wrapped or unwrapped

3 hours at 285° F (140° C), wrapped or unwrapped

6 hours at 250° F (121° C), wrapped or unwrapped

Biologic Testing of the Dry Heat Sterilizer Biologic indicators with endospores of *atrophaeus* are used to monitor the dry heat process. Commercially prepared endospore strips in glassine envelopes should be used. Each load should be tested and the items quarantined until negative results are confirmed.

Chemical Sterilization

The only chemicals used for sterilization are those that are registered as a sterilant by the EPA. They may be approved for use in either a gaseous, plasma, or liquid state.

Ethylene Oxide Gas Sterilization Ethylene oxide (EO) gas is used to sterilize items that are sensitive to heat or moisture. EO or EtO is a chemical alkylating agent that kills microorganisms (including endospores) by interfering with the normal metabolism of protein and reproductive processes, resulting in cell death. Used in the gaseous state, EO must have direct contact with microorganisms on or in the items to be sterilized. EO is the most regulated sterilization process in the United States. EO is highly flammable and explosive in air and therefore must be used in an explosion-proof sterilizing chamber in a controlled environment. The machine is equipped with alarms and will automatically lock if a malfunction occurs. When handled properly, EO is reliable and safe for sterilization,

but the toxic emissions and residues of EO present health hazards to personnel and patients.

Advantages of EO Gas Sterilization

1. EO gas is an effective substitute agent to use with most items that cannot be sterilized by heat, such as plastics with low melting points.
2. EO gas provides an effective method of sterilization for items that steam and moisture may erode; it is noncorrosive and does not damage items.
3. EO gas completely permeates all porous materials; it does not penetrate metal, glass, and petroleum-based lubricants. EO gas sterilization is not recommended for oils, liquids, and powder (including talc). Solutions in glass ampules can be sterilized in EO because the gas does not penetrate glass. But a glass vial with a rubber stopper must not be put in the sterilizer because the gas will penetrate the rubber and may react with the drugs in solution, causing a potentially harmful chemical reaction.
4. Automatic controls preclude human error by establishing proper levels of pressure, temperature, humidity, and gas concentration. The sterilizer is operated according to the manufacturer's instructions.
5. EO gas leaves no film on items.
6. EO gas sterilization is used extensively in the preparation of commercially available, packaged, presterilized items because packaging materials that prolong storage life can be used.

Disadvantages of EO Gas Sterilization

1. EO gas sterilization is a complicated and long process that is carefully monitored.
2. EO sterilization takes longer than steam sterilization; it is a long, slow process.
3. EO gas requires special, expensive equipment. Gas is somewhat expensive per cycle.
4. Items that absorb EO gas during sterilization, such as rubber, polyethylene, or silicone, require an aeration period (Table 18-4). Air admitted to the sterilizer at the end of the cycle only partially aerates the load.
5. Toxic byproducts can be formed in the presence of moisture droplets during the exposure of some plastics, particularly polyvinyl chloride.
6. Repeated sterilization can increase the concentration of the total EO residues in porous items. These increased levels can be hazardous unless the gas can be dissipated.
7. EO is a vesicant when in contact with skin and mucous membranes. a. Liquid EO may cause serious burns if not removed immediately by thorough washing. b. Gloves made of neoprene, polyvinyl fluoride, nitril or butyl rubber, or other material known to be

impermeable to EO penetration should be worn for handling sterilized packages before aeration. If thick cotton gloves are worn, they should be placed in the aerator between uses. c. Personnel who wear contact lenses, especially soft lenses, should wear protective goggles when working around EO sterilizers to avoid eye irritation.

8. Inhaled EO gas can be irritating to mucous membranes. It is a colorless gas, but its presence is easily detectable by odor. Overexposure causes nasal and throat irritation. Prolonged exposure may result in nausea, vomiting, dizziness, difficulty breathing, and peripheral paralysis. a. Immediately after the completion of each cycle, the sterilizer door should be opened approximately 2 inches and the area cleared of all personnel for 15 minutes before unloading. b. Loading carts should be pulled, not pushed, from the sterilizer to the aerator. Air currents flowing over the load may accumulate a residual gas that could be inhaled.

9. Long-term exposure to EO is known to be a potential occupational carcinogen, causing leukemia. It is also a mutagen,

Causing spontaneous abortion, genetic defects, chromosomal damage, and neurologic dysfunction. a. The standards of the Occupational Safety and Health Administration (OSHA) limit an employee's exposure to ethylene oxide to 1 part per million (ppm) of air averaged over an 8-hour period to an active level of 0.5 ppm, and to a short-term limit of 5 ppm averaged over a 15-minute period. These exposure limits are referred to as permissible exposure limits (PELs). The short-term limit addresses exposure to bursts of gas, such as when opening a sterilizer. Breathing zone sampling is performed daily throughout an 8-hour shift on at least one employee for each job classification of exposed personnel. Passive dosimeter badges are the most popular monitoring devices for personnel. Chromatographs and other types of detectors are used for continuous gas analysis of the environment. b. EO gas is vented from the sterilizer to the outside atmosphere to avoid personnel exposure. Audible and visual alarm systems should be installed to indicate a failure in the ventilation system. Most sterilizers have an exhaust hood over the sterilizer door. c. The sterilizer door has locking and sealing mechanisms. The integrity of the seals is checked regularly. Automatic controls must function properly so the door cannot be opened until the gas is evacuated from the chamber. Types of EO Gas Sterilizers the capacity of EO chambers varies from approximately 2 ft (57.5 L)

in a tabletop size of 12 × 12 × 24 inches (30 × 30 × 60 cm) to very large floor-loading units that are 28 × 67 × 78 inches (72 × 180 × 198 cm). These chambers automatically control gas concentration, temperature, humidity, and time. Most models have vacuum pumps to evacuate air from the chamber and steam ejectors to humidify and increase temperature. Microcomputer controls and digital printouts of cycle parameters provide evidence of proper operation. A purge cycle follows the timed gas-exposure cycle to vent the chamber of airborne residual gas.

Ozone Gas Sterilization

Ozone sterilizes by oxidation, a process that destroys organic and inorganic matter. It penetrates the membrane of cells, causing them to explode. Ozone has been used to purify water since the early twentieth century. Ozone is a metastable gas that can be generated easily from oxygen and water at the point of use. No special outlet or water supply is

required. No special cartridges or canisters of a chemical are needed. This is a low-temperature method of sterilization that has been in use in industry for more than 100 years. Safety of ozone gas exposure is measured in PELs. OSHA has set limits as 0.1 ppm per 8-hour period. The PEL is not to exceed 0.3 ppm/15 minutes. Severe pulmonary edema will result at levels of 0.3 ppm exposure for 30 minutes. Personnel using ozone sterilization do not encounter the gas in its intact state. The gas is converted back to water and oxygen at the end of the cycle before the unit can be opened. There is only one setting for all cycles, so the risk for error in operating the machine is minimized. The process takes approximately 1 hour. The FDA has cleared ozone sterilizers for use on plastic, metal, and some rigid lumens. Cellulose packaging cannot be used. Nonwoven pouches or vented closed container systems are used for instrumentation in ozone sterilizers. Ozone sterilization technology was developed in Canada by TSO3 Inc. and approved for sale in 2005.

The sterizone system, marketed by Skytron, operates just above room temperature at 87.4° F to 97° F (30.8° C-36° C) and is safe for heat-sensitive instrumentation. (More information can be found at www.tso3.com.) The ozone sterilizer works in four phases over a period of 4.5 hours:

- Cold water vapor is humidified. About 75 mL per load. O₂ is electrically charged to create O₃, which is ozone. Ozone has a bluish color.
 - The water vapor and O₃ combine in the chamber. One oxygen atom is released from the O₃ molecule to combine with and deactivate microorganisms.
 - The water vapor, ozone, and free oxygen molecule are vented into a catalytic converter that separates out the water and oxygen (H₂O and O₂) for discharge into the drain and room air.
- Advantages of Ozone Gas Sterilization
- The sterilizer generates its own agent using hospital-grade oxygen, water, and the electrical supply. It is simple and inexpensive to operate.
 - Ozone gas sterilization provides an alternative to EO gas sterilization of many heat- and moisture-sensitive items.
 - Ozone gas sterilization does not affect anodized aluminum, titanium, chromium, silicone, neoprene, and Teflon.
 - Aeration is not necessary; ozone leaves no residue and converts to oxygen in a short time.
 - Low temperature is safer for heat-sensitive instrumentation.
- Disadvantages of Ozone

Chemical Sterilants in Solution Liquid

Chemical agents registered as sterilants by the EPA provide an alternative method for sterilizing minimally invasive heat-sensitive items if a gas or plasma sterilizer is not available or the aeration period makes EO gas sterilization impractical. Items processed for

patient care are categorized as critical, semicritical, and noncritical according to the Spaulding classification system. Items that enter tissue or the vascular system are considered critical and must be sterile. Items are sterilized with a liquid sterilant by immersing them in solution for the required time specified by the manufacturer to be sporicidal (i.e., to kill endospores). All chemical solutions have advantages and disadvantages, and each sterilant has specific assets and limitations. (Additional information is available at www.fda.gov.) Policies and procedures should be in place to regulate and guide the use of chemical sterilants in solution. PPE should be worn and eyewash stations should be conveniently located near the point of use. Advantages of Chemical Sterilants

The solution has a low surface tension; it penetrates into crevices and is readily rinsed from items.

It is noncorrosive, nonstaining, and safe for instruments that can be immersed in a chemical solution.

It does not damage lenses or cement on lensed endoscopes.

It is not absorbed by rubber or plastic.

It has low volatility and is stable for the time specified by the manufacturer. Disadvantages of Chemical Sterilants

Prolonged exposure to a chemical sterilant may be necessary for sterilization.

Some chemical sterilants have hazardous effects associated with exposure.

Even if a chemical has low toxicity and irritation, items must be thoroughly rinsed in sterile distilled water before use.

Failure to adequately rinse an endoscope can cause a chemically induced colitis.

Sterile transfer is difficult, because items are wet.

Chemically sterilized items cannot be held in long-term sterile storage. They must be processed for immediate use.

The solution can become diluted during use if an item is wet when placed in it. Types of Chemical Sterilants In addition to EPA registration, chemical sterilants are approved by the U.S. Food and Drug Administration (FDA) as a method of sterilization for critical items that are heat sensitive and can be immersed. The manufacturer is responsible for providing processing instructions on the container label, and the user is obligated to follow the instructions. All solutions should be tested before use with an appropriate test strip to validate the concentration of the sterilant. This is the only way to know if the chemical will be effective.

Acetic Acid. Acetic acid mixed with peracetic acid and hydrogen peroxide kills microorganisms by a process of oxidation to denature proteins in a process that takes 5 to

6 hours at a room temperature of 77° F (25° C). The solution is commonly used as a sterilant for dialysis machines. It is biodegradable and leaves no residue.

Formaldehyde. A 37% aqueous solution (formalin) or 8% formaldehyde in 70% isopropyl alcohol kills microorganisms by coagulating protein in the cells. The solution is effective at room temperature. Formaldehyde has a pungent odor and is irritating to the eyes and nasal passages. Its vapors can be toxic.

Glutaraldehyde. According to the FDA, a 2.4%, 2.5%, or 3.4% aqueous solution of activated buffered alkaline glutaraldehyde kills microorganisms by the denaturation of protein in cells. The solution is activated by adding a powdered buffer to the liquid. Alkaline glutaraldehyde solution changes pH and gradually loses its effectiveness after the date of activation. The expiration date specified by the manufacturer is marked on the container when activated (e.g., 14 or 28 days for aqueous Cidex–activated dialdehyde solution)..

Peracetic Acid.

A proprietary chemical formulation of 35% peracetic acid, hydrogen peroxide, and water inactivates critical microbial cell systems. Peracetic acid is an acetic acid plus an extra oxygen atom that reacts with most cellular components to cause cell death. The mechanism may vary with each type of cell (e.g., vegetative bacterial endospores, mycobacterium). The sterilant is supplied in unit doses for each cycle and is diluted during the sterilization process to 0.2% peracetic acid solution. During the 23-minute sterilization process, the solution is heated to 122° F to 131° F (50° C-55° C) as it passes through the self-contained processing chamber. All items and internal components of the STERIS 1E unit are submerged in the heated sterilant. On completion of the sterilizing cycle, the sterilant is discharged into the sanitary drain. The used chemical is not considered a hazardous material by the EPA. The instruments are automatically rinsed in tap water that is passed through ultraviolet light, filtered through two external prefilters and two 0.1-mm internal microfiltration systems. The smallest known bacterium, *Pseudomonas diminuta*, is unable to pass through the pores in this filter system. (This is the same method used by pharmaceutical manufacturers to make sterile injectable medications.) The STERIS 1E unit uses a standard tap water supply, a sanitary drain, and a 110-V electrical connection. This tabletop unit has a printout to document each cycle. Periodic maintenance includes filter changes based on the chemical components of the external water supply. Biologic monitoring according to the manufacturer's recommendation is performed daily with a commercially prepared endospore strip containing *G. stearothersophilus*. Hypochlorous Acid. Hypochlorous acid is derived from electrochemical activation of a brine solution. Although technically a high-level disinfectant (HLD), it kills many endospores on wellcleaned endoscopes and other heat-sensitive items. This technique was approved in late 2002 by the FDA. Sterilox liquid chemical HLD system is nontoxic and environmentally safe. This chemical is a nonaldehyde oxidation material that kills endospores. The 10-minute process is a single-use design that requires a base unit with inflows of water and electricity that connects to an endoscope processor. Salt tablets (400 grain) and a water conditioner are added at the beginning of the cycle. No special handling or precautions are required. Containers for Chemical Sterilant in Solution A large bin with a perforated inner tray (nested) and lid is used with chemical sterilant solutions at room temperature. With chemical sterilization, items are completely immersed and the lumens filled with sterilant solution. The outside of the bin is labeled with the product name, date

of activation, date of expiration, and initials of the person who mixed the solution. The inside of the bin, inner tray, and its lid should be kept sterile throughout the duration of use of that batch of solution. The lid should remain on the bin to prevent evaporation of the solution and to minimize vaporization of the solution. Evaporation changes the concentration of the mix and will alter the sterilant properties. Formaldehyde and glutaraldehyde solutions are used in a well-ventilated room. Preparing Items for Sterilization by Chemical Immersion Items should be clean and free of organic debris and blood. Items should be washed thoroughly in a nonfilming soapy solution, rinsed, and thoroughly dried before placing them in the chemical sterilant. Lumens should be dried with jets of forced air. The solution will become diluted and lose effectiveness if other solutions Instruments and equipment used in critical areas of the body must be sterile before use. This means the instruments must be completely free of all forms of microorganisms. Common methods of sterilization in the patient health care setting include: • High-temperature steam • **Ethylene oxide (EO) gas** • **Gas plasma sterilization** • **Peracetic acid processing** • **Ozone** • **Dry heat** • **Ionizing radiation** (used in commercial manufacturing of sterile goods). This method is used outside the health care facility and commonly used for bulk sterilization of equipment and supplies.

SELECTING A METHOD

In the past, surgical technologists and nurses have been confronted with decisions about which sterilization method to use for items such as wooden tongue depressors, petroleum-based gauze, liquid in glass bottles, dry powders, and even surgical gloves. These products are now safely sterilized by the manufacturer using technology such as cobalt-60 ionizing radiation. The selection criteria for the modern perioperative environment have changed with advances in technology and economics and with the increased focus on patient and personnel safety. Whereas in the past “instrument nurses” were relied on to choose the correct sterilization method for a given item, now the responsibility has been given over to the manufacturer. The enormous variety of instruments and equipment used in surgery does not fit easily into a generalized protocol for selecting a sterilization method. This complexity of methods also means that the manufacturer’s recommendations must be followed exactly. These methods can also change quickly when new models are marketed. For example, some endoscopic instruments must be hand-cleaned and gas-sterilized, whereas others, even those with digital imaging capability, can be safely sterilized by high pressure steam.

Parameters for sterilization

A method of sterilization is effective only when certain conditions are met. Microbial destruction depends on the individual phases of the process itself and on the bioburden, virulence, and resistance of the microorganisms.

Time: Items must be exposed to the process long enough to destroy microorganisms. • **Saturation and surface exposure:** All parts of the item must be exposed to the sterilization process.

Temperature: Some processes depend on the temperature to which the items are exposed.

Quality of the sterilant, air, and water: The agent must be pure, as must the air and water used in conjunction with it. Chemical or elemental residue in water can reduce the efficacy of the process.

Presence of moisture: Humidity is necessary in most sterilization processes.

Method of packing (where applicable) and loading the sterilizer: This is related to saturation and exposure. Instruments and supplies that have multiple parts or complex design may require disassembly before processing and wrapping. Items must be loaded in a manner that allows the sterilant to penetrate all surfaces. The choice of packaging materials must be suitable for the process.

The following bacteria are used according to the sterilization process:

- Steam sterilization: *Geobacillus stearothermophilus*
- Dry heat and EO sterilization: *Bacillus subtilis*
- Hydrogen peroxide gas plasma: *Bacillus subtilis*
- Peracetic acid: According to the manufacturer's specifications
- High-speed steam sterilizer: *Geobacillus stearothermophilus* enzyme (fluorescence testing)
- Ozone sterilization: *Geobacillus stearothermophilus*

The test bacteria are encased in a self-contained unit that usually is placed in the most challenging area of the load—that is, near the outlet drain of the chamber (steam sterilizer). For EO monitoring, the biological test pack is positioned in the center of the load. After the sterilization process, the bacterial spores are cultured. The growth of bacteria in the test culture indicates that the sterilization process probably was ineffective. If no bacteria grow, all items in the load are presumed to be sterile. Biological controls should be administered at least once weekly in all sterilizers and daily if possible. Biological monitors are always used when an artificial implant or prosthesis is sterilized. If any biological indicator shows a positive result, all items included in that load are withdrawn from use. The infection control department will be notified and the event documented. If the items have already been used in patient treatment, additional safety precautions are immediately implemented, including notification of the surgeon or other primary health care provider who used the items during patient care. Figure 11-13 shows examples of biological monitors. Rapid biological monitoring uses an enzyme that binds to spores. After the sterilization process, the level of enzyme is measured. This corresponds with destruction of the bacterial spores. The monitoring system is used for both high pressure steam and EO sterilization. Results can be obtained in 1 to 4 hours, depending on the sterilization process.

Air Detection Testing To test and monitor the efficiency of the high-vacuum steam sterilizer, a test called the daily air removal test (DART) is performed. Bowie-Dick tests also are commercially available for this purpose. High-vacuum sterilizers are monitored to detect air in the chamber during the exposure phase. In these tests, a special package of properly wrapped towels is taped with heat-sensitive chemical monitor tape and stacked to a height of 10 or 11 inches (25 to 27.5 cm). The package is then placed by itself in the sterilization chamber and run for the appropriate time. An unsatisfactory DART indicates a failure in the vacuum pump system or a defect in the gasket of the sterilizer door. Unsatisfactory results must be reported to the biomedical engineering staff so that they can inspect the sterilizer, especially the vacuum system and door seals.

Steam sterilization

Steam sterilization is the most widely used, effective, and efficient method of sterilization in the health care setting. Normal atmospheric steam is not hot enough to completely

destroy microbes and spores. However, steam under pressure reaches the extremely high temperatures necessary for sterilization. Steam under pressure coagulates the nucleic acids and protein that make up the cell's genetic and enzymatic material. Pressurized steam also destroys the cell's resilient outer wall and bacterial spores. Steam sterilization is achieved according to the temperature, pressure, and exposure time. These depend on the type of steam sterilizer, the size of the load, the temperature, whether items are wrapped, and the type of materials or supplies being processed. The entire process of heating, pressurizing, and timing the load is called a cycle. Modern steam sterilizers have preprogrammed cycles for these parameters. A steam sterilizer has a central chamber where goods are placed and a mechanism for creating extremely high pressure. Steam enters the chamber, air is removed, and the internal pressure increases. This process heats the steam to the very high temperatures required for sterilization. The full cycle may differ slightly among models and manufacturers. However, there are three distinct phases in all types of steam sterilizers:

1. Conditioning (sometimes called preconditioning): Air is removed from the chamber and replaced with steam.

2. Exposure or holding time: Goods in the chamber are exposed to superheated steam at a precise temperature and duration. This duration depends on the size of the load, complexity of materials being sterilized, and whether or not the items were wrapped before sterilization.

3. Exhaust and drying: Pressure in the chamber is reduced and the load is exposed to cool air. Two types of steam sterilizers are used in the clinical setting, gravity displacement and dynamic (also called prevacuum or high-vacuum sterilizers). The main difference in these types is the actual process of removing air and replacing it with steam. The gravity displacement system injects steam into the chamber and displaces the air, which escapes by gravity (downward) through a drain. Dynamic air removal sterilizers use a high-vacuum system to quickly and forcefully evacuate air from the chamber and replace it with bursts or pulses of steam. Gravity Displacement Sterilizer The gravity displacement sterilizer (also called a downward displacement sterilizer) operates on the principle that air is heavier than steam.

All the air must be removed from the inner chamber, because all surfaces of the items must be exposed to the pressurized steam to ensure sterilization. Any air in the inner chamber blocks the passage of pressurized steam to the goods and prevents sterilization. The sterilizer is constructed in such a way that air is displaced downward by gravity (hence the name gravity displacement sterilizer) and is replaced by steam. Dynamic Air Removal (Prevacuum) Sterilizer The dynamic air removal (prevacuum) sterilizer pulls air from the chamber with vacuum force and replaces it with steam. As air is pumped into the chamber, steam enters in pulsed phases. Air is completely removed with positive- and negative-pressure pulses. As the pressure increases, so does the temperature of the steam inside the sterilizer. Superheated steam is possible only with extremely high pressure. The chamber pressure and temperature are held according to the specified program, then the steam is removed through a filter, and the chamber pressure is reduced to normal (atmospheric) temperature. A cool cycle reduces the temperature to a level at which personnel may safely handle the sterilized goods.

Selection of Items for Steam Sterilization Stainless steel instruments, other items made of stainless steel, and woven materials are routinely sterilized with steam. However, many items used in surgery cannot tolerate high pressure, high temperature, and exposure to steam. The appropriate processing method should always be verified directly with the manufacturer's manuals or company representative. Many power-driven instruments and those that contain an optical system or microprocessor cannot be steam-sterilized. Some synthetic materials, such as Silastic, Teflon, polyethylene, polypropylene, and other complex polymers, may also be altered during steam sterilization.

Moisture Concentration (Steam Quality) Effective steam sterilization requires a specific concentration of moisture. If too little moisture is present, items become superheated and eventually can be burned. Too much moisture leaves items wet after removal from the chamber; this can result in contamination of the item. The amount of moisture in the steam is referred to as the steam quality. Water is converted to steam at 212° F (100° C). At this temperature, steam is ineffective for sterilization. Steam that contains more than 97% water is necessary for sterilization to be achieved. Water Quality although the water used during steam sterilization is filtered, contaminants are still present. These usually consist of mineral particles that are too small to be trapped by filters. These minerals are deposited on the surfaces of sterilized items and on the internal surfaces of the sterilizers. The presence of orange, white, brown, or black spots on sterilized items may indicate the presence of excess minerals in the water supply. As these minerals become deposited in the box locks of surgical instruments, they can impair the function of these instruments.

Loading and Operation of a Large-Capacity Steam Sterilizer Large steam sterilizers are used for bulk processing of hospital equipment, including surgical instruments, basins, linen, drapes, and towels. They usually are located in the Central Processing department. Loading techniques for any sterilizer influence the outcome. Large sterilizers used for bulk linens, basins, and complex instruments multiply the problems of air pockets, overheating, and proper steam drainage. Careful loading of items in the steam sterilizer is critical, because if the load is too dense or improperly positioned, air may become trapped in pockets. Items in these air pockets will not be sterilized.

Instrument pans with mesh or wire bottoms are placed flat on the sterilizer shelf.

Linen packs, because of their density, require special attention. Linen packs are best sterilized by placing the packs on their sides.

Packs and instrument trays should be placed so that they do not touch, or touch only loosely, and small items should be placed crosswise over each other.

Heavy packs should be placed at the periphery of the load, where steam enters the chamber.

Basins, jars, cups, or other containers should be placed on their sides with the lids slightly ajar so that the air can flow out of them and steam can enter.

Any item with a smooth surface on which water can collect and drip during the cooling phase of the sterilization cycle should be placed at the bottom of the load.

Modern steam sterilizers are preprogrammed for specific cycles that can be selected by the operator. Information about each load is stored in memory, and hard copies are maintained

for quality assurance. The total time necessary to expose goods to pressurized steam and sterilize them depends on the density of the goods and the temperature in the

Flash (immediate-use) sterilizer

A high-speed prevacuum sterilizer is used in the operating room and in other areas of the hospital to sterilize items for immediate use. Steam sterilization using an abbreviated cycle at a high temperature is necessary for rapid sterilization. However, advanced technology leading to new materials and complex equipment has led to equally complex guidelines for ensuring patient safety. In the past, the term flash sterilization was used to describe the process of a shortened sterilization cycle for uncovered items. Now, however, there are situations in which items are covered in a variety of ways, even for immediate use following sterilization. A standards committee composed of members of accreditation bodies, research professionals, and key health professionals agreed in 2010 that the term flash sterilization does not reflect the complexity of sterilization for immediate use. The term has been replaced with immediate-use sterilization. Standards for immediate-use sterilization have been adapted by the Association of Surgical Technologists as well as other professional organizations.

Ethylene oxide sterilization

Ethylene oxide (EO) is used to sterilize objects that cannot tolerate the heat, moisture, and pressure of steam sterilization. It can be used for delicate instruments, including microsurgical instruments and those with optical systems. It is highly penetrating, which makes it ideal for complex instruments with delicate inner components. EO is a highly flammable liquid that when blended with inert gas, produces effective sterilization by destroying the deoxyribonucleic acid (DNA) and protein structure of microorganisms. EO is used in a 100% pure form, blended with carbon dioxide gas, or mixed with hydrochlorofluorocarbons. EO kills microorganisms and their spores by interfering with the metabolic and reproductive processes of the cell. The process is intensified with both heat and moisture. The EO sterilizer operates at a low temperature. The temperature of the gas directly affects the penetration of items in the chamber. Operating temperatures range from 85° to 100° F (29° to 38° C) for a “cold” cycle and 130° to 145° F (54° to 63° C) for a “warm” cycle. Humidity is required to make spores less resistant to the EO gas. The preferred moisture content for EO sterilization is 25% to 80%. The length of exposure depends on the type and density of material to be sterilized, temperature, humidity, and concentration of the gas. Precautions and Hazards A number of environmental and safety hazards are associated with EO. Exposure to the gas can cause burns to skin and mucous membranes. Prolonged occupational exposure to EO gas is known to cause respiratory damage. EO is a stable compound with a long half-life. Its advantage as a penetrating sterilant also poses hazards. It is difficult to remove, and the level of EO residue is not the same in different materials. In addition, it reacts with many different substances, causing them to disintegrate or creating other toxic compounds, such as ethylene glycol and ethylene chlorohydrins. These chemicals also are difficult to remove from the environment. To prevent injury from EO exposure, the Occupational Safety and Health Administration (OSHA) requires air sampling in areas likely to contain high concentrations of EO. The level of EO must not exceed 0.5 ppm in any area. Personal dosimeter badges measure EO exposure. Air samples should be taken regularly in the sterilization area. Other safety precautions include the installation of an exhaust system that vents the gas to the outside

of the building through an exhaust vent located above the chamber door. Six to 10 exchanges of fresh air per hour are delivered to the sterilization area. Observing the following precautions reduces occupational and patient risk:

After the sterilization process is finished, keep the sterilizer door opened slightly for about 15 minutes.

Make sure goods transported from the sterilizer to the aerator (discussed later) remain on a transport cart.

Always wear protective gloves when handling unaerated items.

Never retrieve an instrument from the aerator before completion of the approved aeration period. Do not ask anyone else to do this, even if a surgeon requests it.

Always pull the transport cart rather than pushing it. Pushing the cart places personnel in back of the air flowing from the unaerated items.

Follow the manufacturer's exact specifications for aeration. Preparation of Items for Gas

Sterilization all items to be gas-sterilized must be clean and dry. Any water left on equipment bonds with EO gas and produces a toxic residue, which can cause burns or a toxic reaction in those who come in contact with it. Any organic material or soil exposed to EO also may produce toxic residues. Therefore, all items processed for EO gas must be completely clean. Items are loaded in the sterilizer loosely so that the gas is free to circulate over every surface. Every attempt should be made to load items that have similar aeration requirements. Some items must not be gas-sterilized.

Self-assessment question

What three conditions are necessary for any sterilization or disinfection process to be effective?

Why are instruments processed in the washer decontaminator sterilizer?

What is the purpose of wrapping instruments for steam sterilization?

What disinfectant is approved for cleaning blood spills in the surgical suite?

Why doesn't a control monitor determine that sterilized goods are sterile?

What is DART?

Describe each of the three levels of the Spaulding classification system.

What is immediate-use processing?

What is terminal disinfection?

Why are sterile packages given a lot number?

Case studies

Case 1

While scrubbed, you open a basin set and find a small amount of water pooled in the bottom of the basin. What are the possible causes of this? What will you do?

Case 2

During the setup for a case, you notice that one of the indicator strips in a pan of instruments on your back table has not changed color. What should you do?

Case 3

You are employed in a small community hospital. In your first week, you are scrubbed on a laparotomy case. The circulator opens a single-use electrosurgery pencil that has been resterilized with EO in the hospital's Central Processing department. What will you do?

Case 4

You are the afternoon shift relief scrub on a plastic surgery case. About an hour into the case on which you have relieved, you happen to notice the chemical sterilization monitor at the bottom of the instrument tray, which is on the sterile back table. The monitor has not changed color. The surgeon is using instruments from this tray. What do you do?

Case 5 You are about to scrub on an orthopedic case in which a stainless steel implant will be inserted into the patient's hip. The surgeon wants to see the implant before surgery. He opens the package and examines it with bare hands. He then instructs you to flash-sterilize the implant. What will you do? Who is responsible? What are the risks associated with flash sterilization of implants?

Case 6 You have been assigned to work in the endoscopy room for the day. You will use gastrointestinal endoscopes. When you arrive in the laboratory, you notice that the endoscope to be used first is soaking in disinfectant solution that has expired (has been held beyond the date approved for disinfection). The physician tells you that he will use the endoscope anyway. What is your response?

Case 7 While opening sterile goods for a case, you find a basin set that has been wrapped in one double-thickness wrapper. You are able to open the basin set using sterile technique. Is it sterile?

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Learning outcome 4: Store Operation Theatre instruments and apparatus

Storage of wrapped sterile goods

After an item has been decontaminated, sorted, wrapped, and sterilized, consideration must be given to how the item can be kept sterile. Time-related sterility measures (based on the length of time since sterilization) are not considered valid. Event-related sterility or terminal sterilization is the accepted standard. Event-related sterility or terminal sterilization is based on the principle that sterilized items are assumed sterile between uses unless environmental conditions or events interfere with the integrity of the package. Guidelines for Storage Items should be stored according to the following guidelines:

Sterile items should be stored in areas that are separate from those used to store clean nonsterile items.

Sterile items must never be stored near sinks or other areas where they can be exposed to water. Sterile items should be stored in critical areas or in the sterile core when possible. They should be placed in a draft free area away from vents and windows. The area must be dust- and lint- free.

Closed cabinets are preferred to open storage areas.

If items are stored in open bins, the bins or drawers should be shallow to prevent excess handling of the items.

Mesh or basket containers are preferable to those with a solid surface where dust and bacteria can collect.

Packages should be placed loosely on shelves to prevent crushing, tearing, or damage to items and wrappers.

Heavy items must never be stacked on top of lighter ones. Stacking heavy instrument trays poses a risk that wrappers will be torn as the top tray is removed.

Seldom-used items can be wrapped in protective dust covers.

Wrappers should be inspected for damage and expiration before they are opened for use

Items commercially prepared and sterilized by manufacturers may be considered sterile indefinitely as long as the wrapper is intact. An expiration date printed on the package shows the maximum time the manufacturer can guarantee product stability and sterility.

Generalized Safety Factors for the Use of Electrosurgery

Electrical burn through the patient's skin is the greatest hazard of electrosurgery. These burns are usually deeper than flame burns, causing widespread tissue necrosis and deep thrombosis to the extent that debridement and grafting may be required. Not all deep thermal injury is immediately apparent. In addition to the precautions noted for preparing the ESU and for positioning the dispersive return electrode used with monopolar units, other precautions should be taken as follows:

1. Electrosurgery (monopolar and bipolar) should not be used in the mouth, trachea, around the head, or in the pleural cavity when high concentrations of oxygen or nitrous oxide are used. During some procedures such as eye surgery, oxygen is administered via nasal cannula or mask. Oxygen builds up under the drapes and sets the stage for a combustible situation. Flame-retardant drapes can conceal a fire in a confined space.

The anesthesia provider is advised to administer medical air instead of oxygen to minimize the risk of fire. Safety regulations for use with all inhalation anesthetic agents are followed. Electrocardiogram electrodes should be placed as far away from the surgical site as possible. Burns can occur at the site of electrocardiogram electrodes and other low-impedance points from invasive monitor probes if current diverts to alternate paths of least resistance. Metallic electromyographic or neuro monitoring needles should not be used.

Rings and other jewelry should be removed. Metallic jewelry, including those used in body piercings, presents a potential risk of burn for the patient from diverted currents from the monopolar unit with either an isolated or ground-referenced output.

Flammable agents such as alcohol should be used with great care in skin preparation. If they are used, the skin surface should be completely dry before draping.

Volatile fumes and vapors may collect in drapes and ignite when the electrosurgical or cautery unit is used.

If another piece of electrical equipment is used in direct contact with the patient at the same time as the ESU, connect it to a different source of current if possible. The cutting current of the ESU may not work if another piece of electrical equipment is on the same circuit. The ESU may interfere with the operation of some equipment, such as older models of cardiac monitors. The isolated power system of solid-state generators may prevent these problems.

New models of cardiac pacemakers are unaffected by monopolar ESU generators. Other implanted electrical devices such as defibrillators could malfunction. Check with the device manufacturer regarding compatibility. The bipolar ESU may be used because the current does not pass through the patient's body and return to the generator. The patient is continuously monitored. A defibrillator should be on standby in the OR.

Connection of a bipolar active electrode to a monopolar receptacle may activate current, causing a short circuit. Plugs on cords should be differentiated to prevent misconnections of active and inactive electrodes. 8. Secure the active electrode handle in an insulated holster/ container when not in use. Do not immerse an active electrode in liquid.

To prevent fire, only moist sponges should be permitted on the sterile field while the ESU is in use. This includes using moist sponges during the use of battery-operated electrocautery. Dry sponges can ignite.

Investigate a repeated request by the surgeon for more current. The dispersive return electrode or connecting cord may be at fault and should be checked first, followed by the handpiece connection. Shock to those touching the patient may result. The patient may be burned at the dispersive return electrode site.

For safety of the patient and personnel, follow instructions for use and care; these appear on the machine or in the manual provided by the manufacturer that accompanies each ESU. Grasp and pull only the plugs, not cords, when disconnecting attachments from the generator or the power source. Position the power cord away from the team to avoid tripping team members. Avoid rolling equipment over the power cord. Disposable cords should not be cut with scissors.

Any malfunctioning ESU should be labeled with the problem and taken out of service until cleared for use by biomedical engineering department personnel.

The patient and personnel should be protected from inhaling plume (smoke) generated during electrosurgery. A suction evacuator device should be placed as close to the source of plume as possible to maximize evacuation of smoke and enhance visibility at the surgical site.

Inspect insulated instruments for breaks in the insulation covering. Current can leak from fractures in the insulation and create a thermal burn.

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Learning outcome 5: Store OT instruments and apparatus

This learning outcome

Storage of Operation Theatre instruments and apparatus

Maintaining Operation Theatre instruments storage records

Principles of Operation Theatre instruments storages

Operation Theatre Storage methods, techniques and structures